EC400 LSE

Revision Mathematics

Notes

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Contents

Chapter 1

Using the Revision Mathematics Notes

Start Revision Maths by doing the quizzes which cover

- Quiz 1: Introduction to Vectors and Matrices.
- Quiz 2: Vectors
- Quiz 3: Matrices
- Quiz 4: Multivariate calculus
- Quiz 5: Introduction to Topology

The quizzes are diagnostic, designed to tell you where you have gaps. If you have no gaps in the quiz questions which are labelled either essential or useful you can move on directly to preparation for the other parts of EC400.

Quiz 1 covers basic material; it you are completely confident that you know this already skip the quiz. You should do quizzes 2, 3 an 4 online. You will then be able to download a document with answers. Skip quiz 5 unless you are an MRes student or wish to take EC487 (Advanced Microeconomics). EC487 is compulsory for MSc Econometrics and Mathematical Economics students; for other students EC487 is available only with permission of the instructor

If you have difficulty with the quizzes read the relevant part of these notes. These are notes only: many results are stated without proof, although in the more important cases intuition is provided. If you had an unlimited amount of time it would be desirable to both have the intuition and understand the proof. Do not despise intuition. It is always helpful. It is easy to be overwhelmed by the amount and complexity of the maths you meet as a postgraduate student in economics. Trying to develop intuition is very helpful. Can you draw a diagram that shows what is happening? Can you explain in words what is going on?

Given the time pressures you are likely to be under, if you can handle the material in the quizzes it is a better use of your time to move on to preparation for the rest of EC400. In economists' language the marginal product of an hour spent preparing for other parts of EC400 is greater than the marginal product of studying the proofs of the results given here.

If you need additional material you may find helpful:

Sydsaeter, H and P. Hammond, **Essential** Mathematics for Economic Analysis. Prentice Hall.

You can use any edition of Sydsaeter and Hammond. Be careful on the exact title, there is a related book Sydsaeter, K, P. Hammond, A. Seirstad and A. Strom, **Further** Mathematics for Economic Analysis which is not appropriate for revision maths.

Chapters 4, 6 and 7 of Sydsaeter and Hammond cover functions of a single variable, including calculus. Chapters 11 and 12 on multivariate calculus, and 15 and 16 on matrices cover the core material of revision maths.

Simon, C.P. and L. Blume, Mathematics for Economists, Norton

is the core text for maths for microeconomics. Chapters 2 - 5 cover background material on functions of a single variable, including calculus. Chapters 8 - 11 on matrices and 13 and 14 on multivariate calculus cover the core material of revision maths, but at greater depth than the treatment here.

Chapter 2

Vectors

2.1 What is a Vector?

For most purposes economists think of vectors as an array of real numbers $(x_1, x_2.....x_n)$. The set of all n vectors is written as $\mathcal{R}^n$, pronounced "R n". You have to distinguish between row and column vectors when doing matrix and vector algebra. In this book all vectors are column vectors.

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The corresponding row vector is written as:

$$\mathbf{x}' = (x_1, x_2.....x_n).$$

The vector $\mathbf{x}'$ is sometimes pronounced as "x prime". Pure mathematicians work with a more abstract definition of a vector and a vector space. Physicists think of vectors as things that have both size and direction, for example velocity and force.

2.2 Vector Addition and Multiplication

2.2.1 Vector Addition

If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in $\mathcal{R}^n$

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

Figure 2.1 illustrates vector addition with the vectors $\mathbf{x}$ and $\mathbf{y}$ shown nose to tail. The alternative is to show vector addition as a parallelogram as in Figure 2.2.

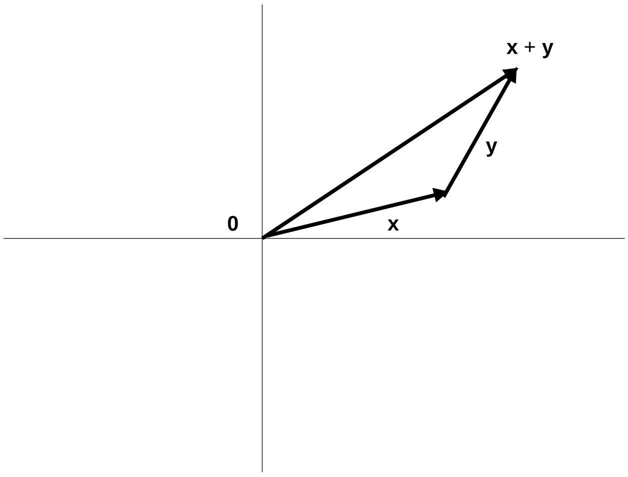

Figure 2.1: Vector Addition

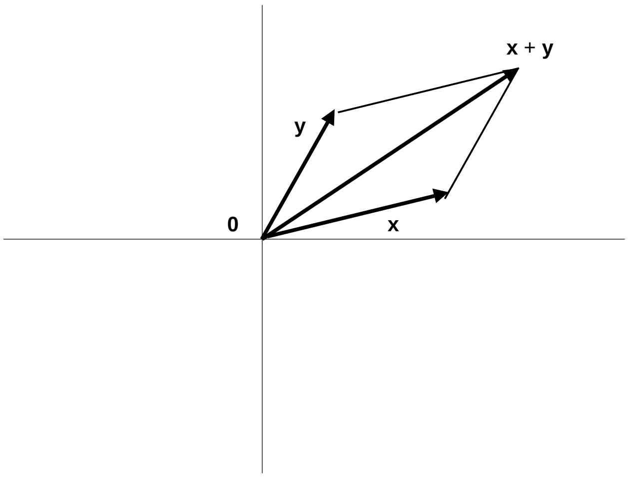

Figure 2.2: Vector Addition Illustrated with a Parallelogram

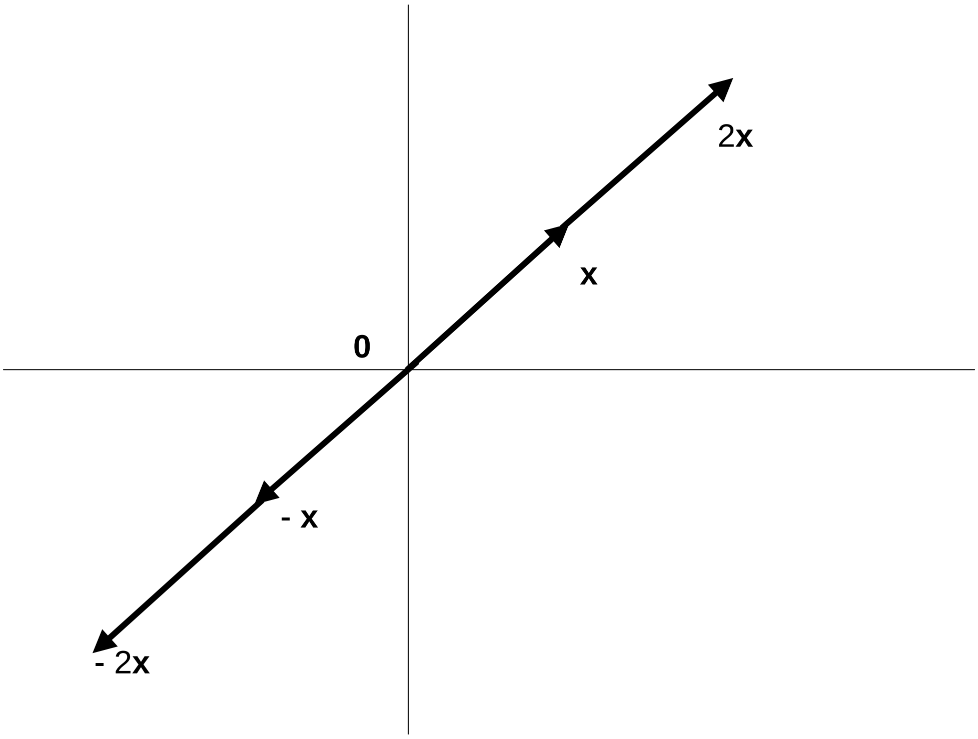

Figure 2.3: Multiplying a Vector by a Scalar

Vector addition is commutative, that is $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$. This is because

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ . \\ . \\ x_n + y_n \end{pmatrix} = \begin{pmatrix} y_1 + x_1 \\ y_2 + x_2 \\ . \\ . \\ y_n + x_n \end{pmatrix} = \mathbf{y} + \mathbf{x}$$

If $\mathbf{x} \in \mathcal{R}^n$ and $\mathbf{y} \in \mathcal{R}^m$ and n and m are different you cannot add $\mathbf{x}$ and $\mathbf{y}$. In mathematical language the sum of $\mathbf{x}$ and $\mathbf{y}$ is undefined.

2.2.2 Scalar Multiplication

If k is a real number (called a scalar in this context)

$$k\mathbf{x} = k \begin{pmatrix} x_1 \\ x_2 \\ : \\ x_n \end{pmatrix} = \begin{pmatrix} kx_1 \\ kx_2 \\ : \\ kx_n \end{pmatrix} = \mathbf{x}k.$$

Figure 2.3 illustrates a vector $\mathbf{x}$ multiplied by 2, -1 and -2. Multiplying a vector by a positive number keeps it pointing in the same direction but makes it longer or shorter. Multiplying a vector by a negative number makes it point in the opposite direction.

Scalar multiplication is distributive over vector addition, that is

$$k(\mathbf{x}+\mathbf{y}) = k\mathbf{x}+k\mathbf{y} \text{ for all } n \text{ vectors } \mathbf{x} \text{ and } \mathbf{y}.$$

To see why observe that

$$\begin{aligned} k\left(\mathbf{x}+\mathbf{y}\right) &= k\begin{pmatrix} x_1+y_1 \\ x_2+y_2 \\ . \\ . \\ x_n+y_n \end{pmatrix} = \begin{pmatrix} k\left(x_1+y_1\right) \\ k\left(x_2+y_2\right) \\ . \\ . \\ k\left(x_n+y_n\right) \end{pmatrix} \\ &= k\begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{pmatrix} + k\begin{pmatrix} y_1 \\ y_2 \\ . \\ . \\ y_n \end{pmatrix} = k\mathbf{x}+k\mathbf{y}. \end{aligned}$$

The definition of vector addition also implies that if h and k are scalars and $\mathbf{x}$ a vector

$$\left(h+k\right)\mathbf{x} = h\mathbf{x}+k\mathbf{x}$$

that is scalar multiplication is distributive over scalar addition.

2.2.3 Inner product

The inner product of two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{R}^n$ is

$$\mathbf{x}'\mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

The inner product is sometime called the "scalar product" or "dot product" and written as $\mathbf{x}\cdot\mathbf{y}$. The properties of the inner product are:

1. The inner product is commutative, that is

   $$\mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x}$$

   for all n vectors $\mathbf{x}$ and $\mathbf{y}$. This is because $\mathbf{x}'\mathbf{y} = \sum_{i=1}^{n} x_i y_i = \sum_{i=1}^{n} y_i x_i = \mathbf{y}'\mathbf{x}$.

2. If $\mathbf{x}$ and $\mathbf{y}$ are n vectors and k is a scalar

   $$k(\mathbf{x}'\mathbf{y}) = (k\mathbf{x})'\mathbf{y} = \mathbf{x}'\left(k\mathbf{y}\right)$$

   because $k(\mathbf{x}'\mathbf{y}) = k\left(\sum_{i=1}^{n} x_i y_i\right) = \sum_{i=1}^{n}(kx_i)y_i = (k\mathbf{x})'\mathbf{y} = \sum_{i=1}^{n} x_i\left(ky_i\right) = \mathbf{x}'\left(k\mathbf{y}\right)$.

3. If $\mathbf{t}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are n vectors

   $$\mathbf{t}'\left(\mathbf{u}+\mathbf{v}\right) = \left(\mathbf{u}+\mathbf{v}\right)'\mathbf{t} = \mathbf{t}'\mathbf{u}+\mathbf{t}'\mathbf{v}$$

   because $\mathbf{t}'\left(\mathbf{u}+\mathbf{v}\right) = \left(\mathbf{u}+\mathbf{v}\right)'\mathbf{t} = \sum_{i=1}^{n} t_i\left(u_i+v_i\right) = \sum_{i=1}^{n} t_i u_i + \sum_{i=1}^{n} t_i v_i = \mathbf{t}'\mathbf{u}+\mathbf{t}'\mathbf{v}$.

4. Also

$$(\mathbf{t}+\mathbf{w})'(\mathbf{u}+\mathbf{v}) = \mathbf{t}'\mathbf{u}+\mathbf{w}'\mathbf{u}+\mathbf{t}'\mathbf{v}+\mathbf{w}'\mathbf{v} = \mathbf{u}'\mathbf{t}+\mathbf{u}'\mathbf{w}+\mathbf{v}'\mathbf{t}+\mathbf{v}'\mathbf{w}$$

because $(\mathbf{t}+\mathbf{w})'(\mathbf{u}+\mathbf{v}) = \sum_{i=1}^{n}(t_i+w_i)(u_i+v_i) = \sum_{i=1}^{n}(t_iu_i+w_iu_i+t_iv_i+w_iv_i)$.

5. Similarly If $\mathbf{t}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are n vectors and a, b, c, d are scalars

$$\begin{aligned}(a\mathbf{t}+b\mathbf{w})'(c\mathbf{u}+d\mathbf{v}) &= (a\mathbf{t})'(c\mathbf{u}+d\mathbf{v})+(b\mathbf{w})'(c\mathbf{u}+d\mathbf{v}) \\ &= (a\mathbf{t})'(c\mathbf{u})+(a\mathbf{t})'(d\mathbf{v})+(b\mathbf{w})'(c\mathbf{u})+(b\mathbf{w})'(d\mathbf{v}) \\ &= ac\mathbf{t}'\mathbf{u}+ad\mathbf{t}'\mathbf{v}+bc\mathbf{w}'\mathbf{u}+bd\mathbf{w}'\mathbf{v}. \qquad (2.1)\end{aligned}$$

2.3 Length and Norm

2.3.1 Definition

For all n vectors $\mathbf{x}$, $\mathbf{x}'\mathbf{x} = \sum_{i=1}^{n} x_i^2$. As $x_i^2 \geq 0$ for all x_i and $x_i^2 = 0$ if and only if $x_i = 0$,

$$\mathbf{x}'\mathbf{x} \geq 0 \text{ for all } \mathbf{x} \text{ and } \mathbf{x}'\mathbf{x} = 0 \text{ if and only if } \mathbf{x} = \mathbf{0}$$

where $\mathbf{0}$ is an n vector with every element 0. Define $\|\mathbf{x}\|$ as

$$\|\mathbf{x}\| = (\mathbf{x}'\mathbf{x})^{\frac{1}{2}}.$$

so $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$. Pythagoras' Theorem implies that in 2 and 3 dimensions $\|\mathbf{x}\|$ is the length of $\mathbf{x}$. Pure mathematicians call $\|\mathbf{x}\|$ the norm of $\mathbf{x}$. It is also called the length of $\mathbf{x}$.

2.3.2 Properties of the Length $\|\mathbf{x}\|$

These are things you need to remember. The first property follows directly from the definition. The others are proved in sections 2.5 and 2.6 of this chapter.

1. If r is a scalar $\|r\mathbf{x}\| = |r|\,\|\mathbf{x}\|$ where $|r|$ is the absolute value of r, so $|r| = r$ if $r \geq 0$ and $|r| = -r$ if $r < 0$.

2. $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\|\,\|\mathbf{y}\|\cos\theta$ where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$.

3. Cauchy-Schwarz inequality:

$$|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\|\,\|\mathbf{y}\|.$$

4. Triangle inequality

$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$
$$\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

5. If $\mathbf{x}'\mathbf{y} = 0$, $\mathbf{x} \neq 0$ and $\mathbf{y} \neq 0$ the angle between $\mathbf{x}$ and $\mathbf{y}$ is $90°$ because $\cos 90° = 0$. The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be orthogonal.

6. If $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\|\,\|\mathbf{y}\|$, $\mathbf{x}$ and $\mathbf{y}$ are parallel and point in the same direction because $\cos 0 = 1$.

7. If $\mathbf{x}'\mathbf{y} = -\|\mathbf{x}\|\,\|\mathbf{y}\|$, $\mathbf{x}$ and $\mathbf{y}$ are parallel and point in opposite directions because $\cos 180° = -1$.

2.4 The Least Squares Problem

2.4.1 The Least Squares Problem With Vector Notation

You will become very familiar with least squares problems in econometrics. This section of notes covers the simplest case. It also generates inequality 2.3 which is an essential step on the way to the Cauchy-Schwarz inequality. The problem here is finding the value of b that minimizes

$$L(b,\mathbf{x},\mathbf{y}) = \sum_{t=1}^{T}(y_t - bx_t)^2.$$

You will come to think of this as the regression of y onto a single variable x without an intercept. Suppose $\mathbf{x}$ and $\mathbf{y}$ are T vectors. Given the definition of the norm $\|\mathbf{y} - b\mathbf{x}\|$

$$L(b,\mathbf{x},\mathbf{y}) = \sum_{t=1}^{T}(y_t - bx_t)^2 = \|\mathbf{y} - b\mathbf{x}\|^2 = (\mathbf{y} - b\mathbf{x})'(\mathbf{y} - b\mathbf{x})$$

Expanding the brackets using the result in equation 2.1 on inner products gives

$$\begin{aligned} L(b,\mathbf{x},\mathbf{y}) &= (\mathbf{y} - b\mathbf{x})'(\mathbf{y} - b\mathbf{x}) \\ &= \mathbf{y}'\mathbf{y} - b\mathbf{x}'\mathbf{y} - b\mathbf{y}'\mathbf{x} + b^2\mathbf{x}'\mathbf{x} \\ &= \mathbf{y}'\mathbf{y} - 2b\mathbf{x}'\mathbf{y} + b^2\mathbf{x}'\mathbf{x} \end{aligned}$$

because $\mathbf{x}'\mathbf{y} = \mathbf{x}'\mathbf{y}$.

Assume that $x_t \neq 0$ for some t so $\mathbf{x} \neq 0$. This implies that $\mathbf{x}'\mathbf{x} > 0$. Then completing the square

$$\begin{aligned} L(b,\mathbf{x},\mathbf{y}) &= \mathbf{y}'\mathbf{y} - 2b\mathbf{x}'\mathbf{y} + b^2\mathbf{x}'\mathbf{x} \qquad (2.2) \\ &= \mathbf{x}'\mathbf{x}\left(b - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})\right)^2 + \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 \\ &\geq \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 \end{aligned}$$

for all b and

$$L(b,\mathbf{x},\mathbf{y}) = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2$$

if and only if $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$. Thus $L(b,\mathbf{x},\mathbf{y})$ is minimized by setting

$$b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$$

and has a minimum value of

$$\mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2.$$

Recall that $L(b,\mathbf{x},\mathbf{y})$ is a sum of squares, so $L(b,\mathbf{x},\mathbf{y}) \geq 0$ for all b including $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$ at which point $L(b,\mathbf{x},\mathbf{y}) = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2$. Thus $\mathbf{y}'\mathbf{y} \geq (\mathbf{x}'\mathbf{x})^{-\mathbf{1}}(\mathbf{x}'\mathbf{y})^{\mathbf{2}}$. Recall that by assumption $\mathbf{x} \neq 0$ so $\mathbf{x}'\mathbf{x} > 0$. Multiplying by $\mathbf{x}'\mathbf{x}$ and rearranging gives

$$(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y}). \qquad (2.3)$$

2.4.2 The Least Squares Problem Without Vector Notation

You can also solve this simple least squares problem without using vectors. The problem is to find b that minimizes the sum of squares.

$$\sum_{t=1}^{n} (y_t - bx_t)^2 .$$

Expanding the brackets

$$\begin{aligned}\sum_{t=1}^{n} (y_t - bx_t)^2 &= \sum_{t=1}^{n} \left(y_t^2 - 2by_tx_t + b^2x_t^2\right) \\ &= \sum_{t=1}^{n} y_t^2 - 2b\left(\sum_{t=1}^{n} x_ty_t\right) + b^2\left(\sum_{t=1}^{n} x_t^2\right).\end{aligned}$$

Assume that at least one of the $x_t's$ is not zero, so $\sum_{t=1}^{n} x_t^2 > 0$. Completing the square

$$\begin{aligned}\sum_{t=1}^{n} (y_t - bx_t)^2 &= \left(\sum_{t=1}^{n} x_t^2\right)\left(b - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)\right)^2 \\ &\quad + \sum_{t=1}^{n} y_t^2 - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)^2\end{aligned}$$

which is minimized by setting

$$b = \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right) = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y}).$$

and has a minimum value of

$$\sum_{t=1}^{n} y_t^2 - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)^2 = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 .$$

Notice that expressions like $\sum_{t=1}^{n} x_ty_t$ take longer to write down and are harder to work with than the corresponding vector expression $\mathbf{x}'\mathbf{y}$. The algebra becomes even worse if you are working with the general least squares problem of minimizing

$$\sum_{t=1}^{n}\left(y_t - \sum_{k=1}^{k} x_{tk}b_k\right)^2 .$$

If you are to survive your econometrics course you must make the switch from the notation $\sum_{t=1}^{n} x_ty_t$ to vector notation $\mathbf{x}'\mathbf{y}$ as rapidly as possible.

2.5 Inequalities for Vectors

2.5.1 The Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality says that for any vectors $\mathbf{x}$ and $\mathbf{y}$ with the same number of elements

$$|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\| \,. \tag{2.4}$$

Here $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ are the norm or length of $\mathbf{x}$ and $\mathbf{y}$, so $\|\mathbf{x}\|^2 = \mathbf{x}'\mathbf{x}$ and $\|\mathbf{y}\|^2 = \mathbf{y}'\mathbf{y}$. The expression $|\mathbf{x}'\mathbf{y}|$ is the absolute value of $\mathbf{x}'\mathbf{y}$. (The absolute value of a number p is p if $p \geq 0$ and $-p$ if $p < 0$. Note that this implies that $p^2 = |p|^2$)

The Cauchy-Schwarz inequality is derived from inequality 2.3 which states that $(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y})$. From the definitions of absolute value and norm $|\mathbf{x}'\mathbf{y}|^2 = (\mathbf{x}'\mathbf{y})^2$, $(\mathbf{x}'\mathbf{x}) = \|\mathbf{x}\|^2$ and $\mathbf{y}'\mathbf{y} = \|\mathbf{y}\|^2$ so the inequality $(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y})$ can be written as

$$|\mathbf{x}'\mathbf{y}|^2 \leq \|\mathbf{x}\|^2 \, \|\mathbf{y}\|^2 \tag{2.5}$$

which implies the Cauchy-Schwarz inequality $|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$. Inequality 2.3 was proved under the assumption that $\mathbf{x} \neq 0$, but as both sides of the Cauchy-Schwarz inequality are zero when $\mathbf{x} = 0$ it also holds when $\mathbf{x} = 0$.

2.5.2 The Triangle Inequality

The triangle inequality states that

$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \,.$$

Geometrically you can think of $\mathbf{x}+\mathbf{y}$, $\mathbf{x}$ and $\mathbf{y}$ as three sides of a triangle. The inequality simply states that one side of a triangle cannot be longer than the sum of the other two sides, as illustrated in Figure 2.4.

The triangle inequality is a direct consequence of the Cauchy-Schwarz inequality. From the definition of $\|\mathbf{x}+\mathbf{y}\|$

$$\|\mathbf{x}+\mathbf{y}\|^2 = (\mathbf{x}+\mathbf{y})'(\mathbf{x}+\mathbf{y}) = \mathbf{x}'\mathbf{x} + \mathbf{2x}'\mathbf{y} + \mathbf{y}'\mathbf{y}$$

From the Cauchy-Schwarz inequality $|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$. As $z \leq |z|$ for any number z this implies that $\mathbf{x}'\mathbf{y} \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$ so

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\|^2 &= \mathbf{x}'\mathbf{x} + \mathbf{2x}'\mathbf{y} + \mathbf{y}'\mathbf{y} \\ &\leq \|\mathbf{x}\|^2 + \mathbf{2}\,\|\mathbf{x}\| \, \|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \,. \end{aligned}$$

As $\|\mathbf{x}+\mathbf{y}\|$, $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ are all non negative this implies the triangle inequality $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. Replacing $\mathbf{y}$ by $-\mathbf{y}$ and noting that $\|\mathbf{y}\| = \|-\mathbf{y}\|$ the triangle inequality implies that

$$\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \,.$$

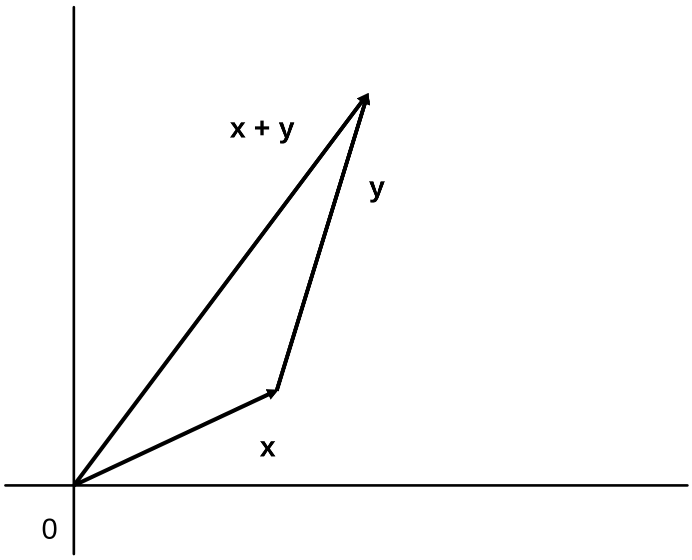

Figure 2.4: triangle inequality

2.6 The Angle Between Two Vectors

2.6.1 Showing That $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\theta$

Think of $\mathbf{x}$ and $\mathbf{y}$ as non zero vectors and b as a scalar. In Figure 2.5 $b\mathbf{x}$ is a scalar multiple of the vector $\mathbf{x}$. The least squares problem is finding b to minimize $\|\mathbf{y}-b\mathbf{x}\|$ which is the distance between the point C at the end of the vector $\mathbf{y}$ and the horizontal straight line from the origin 0 that goes through the end of the vector $\mathbf{x}$. Geometrically you find the point $b\mathbf{x}$ on the line $0\mathbf{x}$ that minimizes the length of $\mathbf{y}-b\mathbf{x}$ by drawing the line from C to B that is at 90° to the vector $\mathbf{x}$..

As $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$ solves the least squares minimization problem the length of the line OB is the length of the vector $b\mathbf{x}$ where $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$. Think about the case where b is positive as shown in Figure 2.5. As $\|\mathbf{x}\| = (\mathbf{x}'\mathbf{x})^{\frac{1}{2}}$ the length of OB is $\|\mathbf{x}\| \, b = \|\mathbf{x}\| (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y}) = (\mathbf{x}'\mathbf{x})^{-\frac{1}{2}}(\mathbf{x}'\mathbf{y})$. Elementary trigonometry and Figure 2.5 imply that if θ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$

$$\cos\theta = \frac{0B}{0C} = \frac{(\mathbf{x}'\mathbf{x})^{-\frac{1}{2}}(\mathbf{x}'\mathbf{y})}{\|\mathbf{y}\|} = \frac{\mathbf{x}'\mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|} \tag{2.6}$$

where $\cos\theta$ is the cosine of the angle θ. Equation 2.6 implies that

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\theta.$$

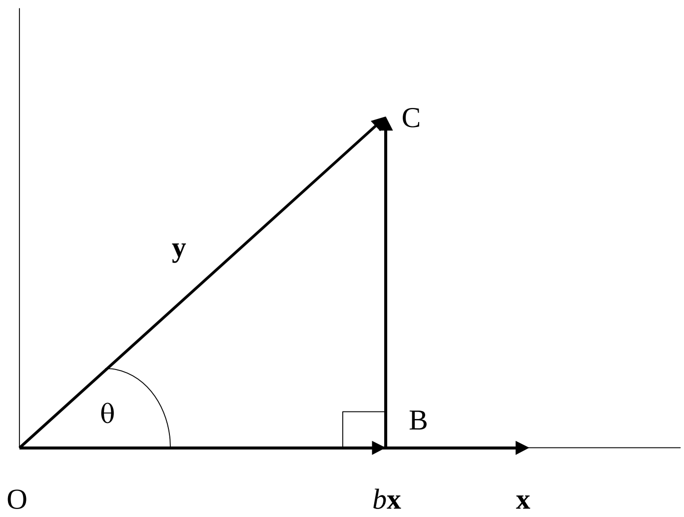

Figure 2.5:

In words this says that the inner product of the vectors **x** and **y** is the product of their lengths and the cosine of the angle between the two vectors. The relationship also holds when b is negative, so the angle between the two vectors is more than $90°$ and, as the graph of $\cos\theta$ in Figure 2.6 shows, $\cos\theta < 0$. One of the properties of the cosine is that for all θ with $0° \leq \theta \leq 360°$

$$\cos\theta = \cos(360 - \theta).$$

This has the useful implication that it does not matter which way round you think of the angle between the two vectors, as shown in Figure 2.7 . If you think of the angle between the two vectors as the smaller of the two angles between the two vectors, so it lies between $0°$ and $180°$. The graph of the function $\cos\theta$ in Figure 2.6 shows that

$$0 < \cos\theta < 1 \text{ if } 0 < \theta < 90°$$

and

$$-1 < \cos\theta < 0 \text{ if } 90° < \theta < 180°$$

which gives a result sufficiently useful to be stated formally.

Proposition 1

- $\mathbf{x}'\mathbf{y} > 0$ *if the angle between* $\mathbf{x}$ *and* $\mathbf{y}$ *is between* 0 *and* $90°$.
- $\mathbf{x}'\mathbf{y} < 0$ *if the angle between* $\mathbf{x}$ *and* $\mathbf{y}$ *is between* $90°$ *and* $180°$.

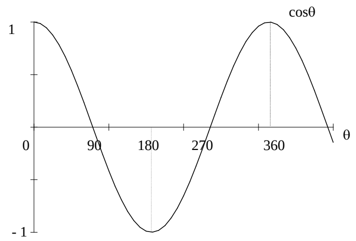

Figure 2.6: The Graph of $\cos\theta$

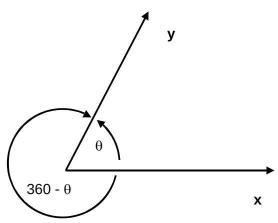

Figure 2.7:

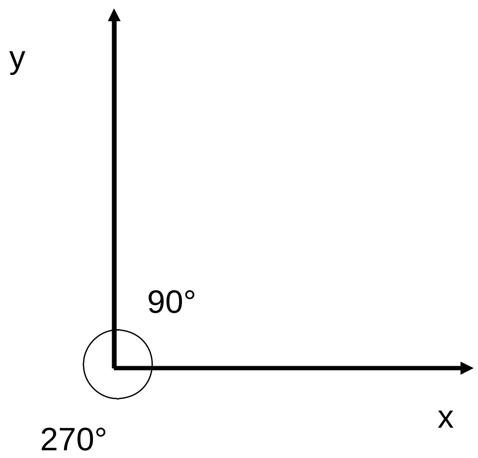

Figure 2.8:

2.6.2 Orthogonal Vectors

Suppose $\mathbf{x}$ and $\mathbf{y}$ are non zero vectors, so $\|\mathbf{x}\| > 0$ and $\|\mathbf{y}\| > 0$. Suppose also that $\mathbf{x}'\mathbf{y} = 0$. This implies that $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = 0$ where θ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$. As $\|\mathbf{x}\| > 0$ and $\|\mathbf{y}\| > 0$ this implies that $\cos\theta = 0$. Given the graph of $\cos\theta$ this implies that $\theta = 90°$ or $\theta = 270°$.

As Figure 2.8 shows you can measure the angle between $\mathbf{x}$ and $\mathbf{y}$ as 90° or 270° . It is convenient to choose the smaller angle 90°. In mathematical language the vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be orthogonal if $\mathbf{x}'\mathbf{y} = 0$ so the vectors are at an angle of 90° to each other. In everyday English an angle of 90° is called a "right angle".

2.6.3 Parallel vectors

If

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = \|\mathbf{x}\| \|\mathbf{y}\|$$

then $\cos\theta = 1$ so $\theta = 0$. The angle between $\mathbf{x}$ and $\mathbf{y}$ is 0°. The two vectors are parallel and point in the same direction.

If

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = -\|\mathbf{x}\| \|\mathbf{y}\|$$

then $\cos\theta = -1$ so $\theta = 180°$. The two vectors are parallel and point in opposite directions.

2.7 Vectors, Lines, Planes and Hyperplanes

2.7.1 Vectors and Lines in $\mathcal{R}^2$

The objective of this section is to introduce the idea of a hyperplane, which is a generalization of a line in two dimensional space, and a plane in three dimensional space. Hyperplanes are important for an intuitive understanding of partial derivatives, and also for the theory of concavity and convexity that underlies optimization theory.

One of microeconomics' favorite equations is the budget line; with two goods x_1 and x_2, with prices p_1 and p_2 this is the set of points satisfying

$$p_1 x_1 + p_2 x_2 = m \tag{2.7}$$

where m is the amount the consumer has to spend. This is straight line; if $p_2 \neq 0$ equation 2.7 can be written as

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1$$

so the slope of the budget line is $-\frac{p_1}{p_2}$. If $p_2 = 0$ the line is vertical. In vector notation equation 2.7 can be written as

$$\mathbf{p}'\mathbf{x} = m.$$

Another way at looking at this equation is to choose any point $\mathbf{x}_0$ on the line so $\mathbf{p}'\mathbf{x}_0 = m$ and write the equation as $\mathbf{p}'\mathbf{x} = \mathbf{p}'\mathbf{x}_0$, or rearranging

$$\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0.$$

Recall that the inner product $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = \|\mathbf{p}\| \|\mathbf{x} - \mathbf{x}_0\| \cos\theta$ where θ is the angle between the two vectors $\mathbf{p}$ and $\mathbf{x} - \mathbf{x}_0$. Thus as $\cos 90° = 0$, if the inner product is 0, the angle is 90° and the two vectors are said to be "orthogonal". The line, that is the set of points satisfying $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0)$, is precisely the set of points for which $\mathbf{x} - \mathbf{x}_0$ is orthogonal to $\mathbf{p}$. This is illustrated in Figure 2.9. The vector $\mathbf{p}$ orthogonal to the line is called the **normal vector**.

2.7.2 Vectors and Planes in $\mathcal{R}^3$

With three goods the budget equation becomes

$$p_1 x_1 + p_2 x_2 + p_3 x_3 = m.$$

The vector notation appears unchanged as $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0$, but now you have to think of the vectors as living in three dimensional space $\mathcal{R}^3$ rather than two dimensional space $\mathcal{R}^2$. Geometrically, in three dimensional space, given a fixed vector $\mathbf{p}$, the set of vectors orthogonal to $\mathbf{p}$ is a plane. The vector $\mathbf{p}$ is again called the **normal vector** to the plane.

You can perhaps imagine the plane and its normal vector $\mathbf{p}$ by thinking of your pen as $\mathbf{p}$. If you put one end of the pen on the table top, and point the pen

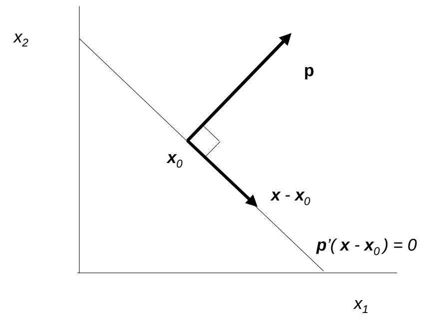

Figure 2.9: The vector $\mathbf{p}$ is orthogonal to the line $\mathbf{p}'(\mathbf{x}-\mathbf{x_0})=0$.

vertically upwards, all the vectors in the plane of the table top are horizontal, so are orthogonal to the pen. It is possible to lift up a corner of the table so the table top is no longer horizontal, whilst keeping the angle between pen and table top fixed at 90°, so the vector is still orthogonal to the plane. (Take your coffee mug off the table before trying this.)

2.7.3 Vectors and Hyperplanes in $\mathcal{R}^n$

With n goods the budget equation becomes

$$p_1x_1 + p_2x_2 + ... + p_nx_n = m.$$

Again the budget equation can be written as $\mathbf{p}'(\mathbf{x}-\mathbf{x}_0)=0$, but $\mathbf{p}$ and $\mathbf{x}-\mathbf{x}_0$ are now n vectors. A hyperplane is defined as follows:

Definition 2 *A hyperplane in $\mathcal{R}^n$ is a set of the form*

$$\{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n, \mathbf{p}'(\mathbf{x}-\mathbf{x}_0)=0\}$$

where $\mathbf{p}$ is a non-zero vector in $\mathcal{R}^n$, and $\mathbf{x}_0$ is a vector in $\mathcal{R}^n$.

In two dimensional space a hyperplane is a straight line, and in three dimensional space a hyperplane is a plane. For any n all the vectors $\mathbf{x}-\mathbf{x}_0$ are orthogonal to $\mathbf{p}$, and $\mathbf{p}$ is still called the normal vector.

Chapter 3

Matrices

3.1 Introduction

This chapter sets out the essential facts about matrices that you have to know as well as you know your route from LSE to home if you are to survive your econometrics course. The starting point is the definition of the basic operations of addition, subtraction and multiplication for matrices. These are the foundations of matrix algebra. I then give the rules of matrix algebra that follow from these definitions. Finally I discuss the circumstances in which a matrix has an inverse, introducing the ideas of linear independence, spanning and rank.

The treatment in this chapter is very cookbook - intuition, not proof. It's intention is to give you quickly the essentials of what you need to know on matrices. But the rules for matrices, in particular matrix multiplication, can seem very arbitrary. I explain why this is not arbitrary in section 3.14 at the end of this chapter. There is no need for you to read this, look at it only if you are interested. I explain that matrices are a natural representation of linear functions, that matrix multiplication is implied by the composition of functions, and the results on the existence of a matrix inverse are implied by the general conditions for the existence of a function inverse. .

3.2 What is a Matrix?

A matrix is a rectangular array of numbers, for example the matrix A is

$$A = \begin{pmatrix} A_{11} & A_{12} & . \; . & A_{1m} \\ A_{21} & A_{22} & . \; . & A_{2m} \\ : & : & : \; : & : \\ A_{n1} & A_{n2} & . \; . & A_{nm} \end{pmatrix}.$$

A has n rows and m columns, and is described as an $n \times m$ matrix. A_{ij} is the element in row i and column j. Assume that all the numbers I mention are real unless I explicitly say that you should consider the possibility that they are complex. (This arises only in the chapter on eigenvalues and eigenvectors.)

3.3 Definitions for Matrix Algebra

You can do algebra with matrices just as you can do algebra with numbers. The first stage in developing matrix algebra is to define addition, subtraction and multiplication with matrices. I am going to ignore division for now, and come back to it later in the form of the matrix inverse A^{-1}. The most important difference between algebra with matrices and algebra with the numbers is that you can do addition, subtraction and multiplication with any pair of numbers. This is not so with matrices so the definitions of these operations have to include a statement on which matrices the operations can be done with.

Addition and subtraction are straightforward.

Definition 3 (Matrix Addition) *Two matrices A and B can be added if they are both $n \times m$ matrices. The sum $A + B$ is defined as the $n \times m$ matrix for which component ij is $A_{ij} + B_{ij}$.*

Definition 4 (Matrix Subtraction) *Two matrix B can be subtracted from the matrix A if both A and B are $n \times m$ matrices. The difference $A - B$ is defined as the $n \times m$ matrix for which component ij is $A_{ij} - B_{ij}$.*

There are two types of multiplication to be considered. The first is multiplication of a matrix by a number, often referred to as a scalar in this context.

Definition 5 (Multiplication of a Matrix by a Number) *Any $n \times m$ matrix A can be multiplied by any number θ. The product θA is defined as the $n \times m$ matrix for which component ij is θA_{ij}.*

It is also possible to multiply some pairs of matrices.

Definition 6 (Multiplication of Two Matrices) *If B is an $n \times m$ matrix and C is a $m \times p$ matrix the product BC is defined as the $n \times p$ matrix for which component ij is*

$$\sum_{h=1}^{m} B_{ih} C_{hj}.$$

3.4 Matrix Algebra 1: When Matrix Algebra is Like Ordinary Algebra

Some of the rules for matrix algebra are the same as the rules for ordinary algebra where the variables are real numbers. These matrix algebra rules follow straight from the definition of matrix addition and multiplication and the corresponding properties of ordinary addition and multiplication.

1. **Commutative Addition**

   If A and B are $n \times m$ matrices

$$A + B = B + A.$$

2. **Associative Addition**

   If A, B and C are all $n \times m$ matrices

$$(A + B) + C = A + (B + C).$$

   so it is possible to write simply $A + B + C$.

3. **Associative Multiplication of a Matrix by a Scalar**

   If λ and μ are numbers and A is a matrix

$$\lambda\,(\mu A) = (\lambda\mu)\,A$$

4. **Distributive Scalar Addition and Multiplication**

   If λ and μ are numbers and A is a matrix

$$(\lambda + \mu)\,A = \lambda A + \mu A.$$

5. **Distributive Matrix Addition and Scalar Multiplication**

   If λ is a number and A and B are $n \times m$ matrices

$$\lambda\,(A + B) = \lambda A + \mu B.$$

6. **Associative Multiplication**

   If A is an $n \times m$ matrix, B a $m \times p$ matrix and C an $p \times q$ matrix

$$(AB)C = A(BC).$$

   so it is possible to write simply ABC.

7. **Distributive Addition and Multiplication**

   If A is an $n \times m$ matrix, B and C $m \times p$ matrices and D an $p \times q$ matrix

$$\begin{aligned} A(B + C) &= AB + AC \\ (B + C)D &= BD + CD. \end{aligned}$$

8. **Expanding Brackets**

   Distributive addition and multiplication imply that if A and B are $n \times m$ matrices and C and D $m \times p$ matrices

$$(A + B)(C + D) = AC + BC + AD + BD.$$

9. **The Zero Matrix**.

   The zero matrix 0 has all elements 0. It behaves like 0 in ordinary algebra.

$$\begin{aligned} A + 0 &= A \\ A0 &= 0. \end{aligned}$$

10. **The Identity Matrix**

    The $n \times n$ identity matrix I^n has $I^n_{ii} = 1$ for $i = 1, 2...n$ and $I^n_{ij} = 0$ if $i \neq j$, so

$$I_n = \begin{pmatrix} 1 & 0 & .. & 0 \\ 0 & 1 & .. & 0 \\ : & : & :: & : \\ 0 & 0 & .. & 1 \end{pmatrix}.$$

    In practice we omit the n, and use I for any identity matrix. The identity matrix behaves like 1 in ordinary algebra.

$$\begin{aligned} IA &= A \\ AI &= A. \end{aligned}$$

3.5 Matrix algebra 2: When Matrix Algebra is Not Like Ordinary Algebra

1. You cannot add many pairs of matrices. You can only add A and B if they have the same number of rows and columns so both A and B are $n \times m$ matrices.

2. You cannot multiply many pairs of matrices. You can only multiply A by B if the number of columns of A is the same as the number of rows of B, so A is a $n \times m$ matrix and B is $m \times p$ matrix.

3. Matrix multiplication is not commutative. There are some matrices A and B for which $AB = BA$, but in general $AB \neq BA$. For example

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 3 & 1 \end{pmatrix}$$

    but

$$\begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 3 & -2 \end{pmatrix}.$$

4. By definition the matrix A^{-1} is the inverse of A if

$$A^{-1}A = AA^{-1} = I.$$

    Many matrices do not have an inverse A matrix cannot have an inverse unless it is square, that is has the same number of rows and columns, and many square matrices do not have an inverse. See section 3.10 of these notes for details.

3.6 The Transpose of a Matrix

The transpose A' of the $n \times m$ matrix A is a $m \times n$ matrix with $A'_{ij} = A_{ji}$. You may see the notation A^T used for the transpose of A. The notation $\mathbf{x}'$ for the row vector with the same elements as the column vector $\mathbf{x}$ simply treats the n vector $\mathbf{x}$ as an $n \times 1$ matrix. Transposes are important in econometrics, for example the ordinary least squares estimator is $\left(X'X\right)^{-1}X'\mathbf{y}$ involves a transpose.

The definition of a transpose implies that

1. $(A+B)'=A'+B'$ because
$$(A+B)'_{ij}=(A+B)_{ji}=A_{ji}+B_{ji}=A'_{ij}+B'_{ij}.$$
2. $(A')'=A$ because $(A'_{ij})'=A'_{ji}=A_{ij}$.
3. $I'=I$ because $I'_{ii}=I_{ii}=1$ and if $j\neq i$ $0=I_{ij}=I_{ji}=I'_{ij}$.
4. If A is an $n\times m$ matrix and B a $m\times p$ matrix
$$(AB)'=B'A'$$
because $(AB)'_{ji}=(AB)_{ij}=\sum_{h=1}^{m}A_{ih}B_{hj}=\sum_{h=1}^{m}A'_{hi}B'_{jh}=\sum_{h=1}^{m}B'_{jh}A'_{hi}=(B'A')_{ji}$.
This implies that if A_1, $A_2,.....A_{m-1},A_m$ are matrices, and A_i an $n_{i-1}\times n_i$ matrix for $i=1,2....m$
$$(A_1A_2....A_{m-1}A_m)'=A'_mA'_{m-1}....A'_2A'_1.$$
5. If the $n\times n$ matrix A has an inverse
$$(A')^{-1}=(A^{-1})'$$
because
$$I=\left(A^{-1}A\right)=I'=\left(A^{-1}A\right)'=A'\left(A^{-1}\right)'$$
and
$$I=\left(AA^{-1}\right)=I'=\left(AA^{-1}\right)'=\left(A^{-1}\right)'A'.$$
so $A'(A^{-1})'=(A^{-1})'A'=I$ so the inverse of A' is $(A^{-1})'$.
6. If B is a 1×1 matrix, that is a scalar $B'=B$. Thus if x is an n vector, A an $n\times m$ matrix and y a m vector, $x'Ay$ is a scalar so
$$x'Ay=(x'Ay)'=y'A'x.$$

3.7 The Trace of a Square Matrix

If A is a square $n\times n$ matrix the trace of A, written as $tr\ A$, is the sum of the diagonal terms, that is
$$trA=\sum_{i=1}^{n}A_{ii}.$$

Matrices that are not square do not have a trace. If B is an $n \times k$ matrix and C is a $k \times n$ matrix, BC is an $n \times n$ matrix and CB is a $k \times k$ so both BC and CB are square matrices. Further

$$tr\,(BC) = tr\,(CB)\,.$$

This is easily proved by noting that

$$tr\,(BC) = \sum_{i=1}^{n} (BC)_{ii} = \sum_{i=1}^{n} \left(\sum_{j=1}^{k} B_{ij} C_{ji} \right) = \sum_{j=1}^{k} \left(\sum_{i=1}^{n} C_{ji} B_{ij} \right) = \sum_{j=1}^{k} (CB)_{jj} = tr\,(CB)\,.$$

Econometricians are particularly fond of the fact that if $\mathbf{x}$ is an n vector, and A an $n \times n$ matrix, then $\mathbf{x}'A\mathbf{x}$ is a scalar, that is a 1×1 matrix. Thus $\mathbf{x}'A\mathbf{x} = tr\,(\mathbf{x}'A\mathbf{x})$ which implies that

$$\mathbf{x}'A\mathbf{x} = tr\,(\mathbf{x}'A\mathbf{x}) = tr\,(A\mathbf{x}\mathbf{x}')\,.$$

3.8 Special Square Matrices

There are a number of special matrices you need to know about.

- The matrix A is **diagonal** if $A_{ij} = 0$ for all $i \neq j$, so all the off diagonal elements are zero. For example the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

  is diagonal, but the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 4 & 3 \end{pmatrix}$$

  is not diagonal.

- The matrix A is **upper triangular** if $A_{ij} = 0$ for all $i < j$, so all the elements below the diagonal are zero. For example the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

  is upper triangular, but the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

  is not upper triangular.

- The matrix A is **lower triangular** if $A_{ij} = 0$ for all $i > j$, so all the elements above the diagonal are zero. For example the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is lower triangular, but the matrix

$$\begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is not lower triangular.

- The matrix A is **symmetric if** $A' = A$ that is $A_{ij} = A_{ji}$ for all i and j. For example the matrix

$$\begin{pmatrix} 1 & 3 & 4 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is symmetric, but the matrix

$$\begin{pmatrix} 1 & 0 & 2 \\ 3 & 2 & 0 \\ 4 & 0 & 3 \end{pmatrix}$$

is not symmetric.

- The matrix A is **orthonormal** if A' is the inverse of A so $A\ A' = I$. For example the matrix

$$\begin{pmatrix} 2^{-\frac{1}{2}} & -2^{-\frac{1}{2}} \\ 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix}$$

is orthonormal because

$$\begin{pmatrix} 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \\ -2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} 2^{-\frac{1}{2}} & -2^{-\frac{1}{2}} \\ 2^{-\frac{1}{2}} & 2^{-\frac{1}{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Using notation A^h for column h of the matrix A, and thinking of A^h as a vector the matrix A is orthonormal if $\left\|A^h\right\| = 1$ for all h, and $\left(A^h\right)' A^k = 0$ for $h \neq k$. This says that the vectors that make up the columns of A all have length 1 and are orthogonal. to each other.

- The matrix A is **idempotent** if $A^2 = A$. (The notation A^2 is used in the obvious way to mean AA.)

  For example the matrix

$$\begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix}$$

is idempotent because

$$\begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ -1 & -2 \end{pmatrix}.$$

The most important example of an idempotent matrix is $X\left(X'X\right)^{-1} X'$.

In Mathematics for Microeconomics you will meet the following definitions.

- The symmetric $n \times n$ matrix A is **positive semidefinite** if $\mathbf{x}'A\mathbf{x} \geq 0$ for all n vectors $\mathbf{x}$.
- The symmetric $n \times n$ matrix A is **positive definite** if $\mathbf{x}'A\mathbf{x} \geq 0$ for all n vectors $\mathbf{x}$ and $\mathbf{x}'A\mathbf{x} = 0$ if and only if $\mathbf{x} = 0$.
- The symmetric $n \times n$ matrix A is **negative semidefinite** if $\mathbf{x}'A\mathbf{x} \leq 0$ for all n vectors $\mathbf{x}$.
- The symmetric $n \times n$ matrix A is **negative definite** if $\mathbf{x}'A\mathbf{x} \leq 0$ for all n vectors $\mathbf{x}$ and $\mathbf{x}'A\mathbf{x} = 0$ if and only if $\mathbf{x} = 0$.

3.9 Inverse Matrices

The matrix A has an inverse A^{-1} if and only if $A^{-1}A = I$ and $AA^{-1} = I$. A matrix that has an inverse is said to be invertible. The critical thing that you must remember are

1. If a matrix has different numbers of rows and columns it cannot have an inverse.
2. If a matrix has the same number of rows and columns it may not have an inverse.
3. A matrix has at most one inverse. To see that suppose that both B and C are inverses of A so $BA = I$ and $AC = I$, then
$$B = BI = B(AC) = (BA)C = IC = C.$$
4. The inverse of A^{-1} is A. This is implied by $A^{-1}A = I$ and $AA^{-1} = I$.
5. If A and B are both $n \times n$ invertible matrices the matrix AB is invertible and
$$(AB)^{-1} = B^{-1}A^{-1}.$$
This is easily proved by noting that $(AB)\left(B^{-1}A^{-1}\right) = ABB^{-1}A^{-1} = A\left(BB^{-1}\right)A^{-1} = AIA^{-1} = AA^{-1} = I$, and similarly $B^{-1}A^{-1}AB = I$.
6. More generally a similar argument implies that if A_1, $A_2, ...A_{m-1}, A_m$ are invertible $n \times n$ matrices, the matrix $A_1A_2....A_{m-1}A_m$ is invertible and
$$(A_1A_2....A_{m-1}A_m)^{-1} = A_m^{-1}A_{m-1}^{-1}....A_2^{-1}A_1^{-1}.$$

3.10 When Does a Matrix Have an Inverse?

There are various ways of approaching the question of whether a matrix has an inverse. A way of getting at this is working with row echelon matrices. This emphasizes methods for finding the inverse of a matrix using pen and paper. It is useful to know how to find the inverse of of a 2×2 matrix. But with

If $A_{11}A_{22} - A_{12}A_{21} = 0$ the matrix

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

has no inverse. If $A_{11}A_{22} - A_{12}A_{21} \neq 0$ the matrix has an inverse

$$A^{-1} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} = (A_{11}A_{22} - A_{12}A_{21})^{-1} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}$$

Think of this as swapping the diagonal elements changing the sign of the off diagonal elements, and dividing by the determinant of the matrix which is $A_{11}A_{22} - A_{12}A_{21}$.

For more general matrices if you are doing algebra you need to think about whether the matrix has an inverse, but if you are confident that the inverse exists just write A^{-1}. On no account should you write out A^{-1} component by component. If you need the numbers use a computer. Econometics packages have a matrix inversion routine built into them.

The big result on matrix inversion is:

Theorem 7 *Only square matrices have an inverse. Not all square matrices have an inverse. If A is a square $n \times n$ matrix either*

- *A has an inverse*
- *the only vector for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$*
- *the columns of A are linearly independent*
- *the columns of A span $\mathcal{R}^n$.*
- *rank $A = n$*
- *the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$ in $\mathcal{R}^n$*
- *the determinant* $\det A \neq 0$

or

- *A has no inverse*
- *there is a vector $\mathbf{x} \neq \mathbf{0}$ for which $A\mathbf{x} = \mathbf{0}$*
- *the columns of A are are not linearly independent*
- *the columns of A do not span $\mathcal{R}^n$.*
- *rank $A < n$*
- *depending on $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ either has no solution or an infinite number of solutions*
- *the determinant* $\det A = 0$.

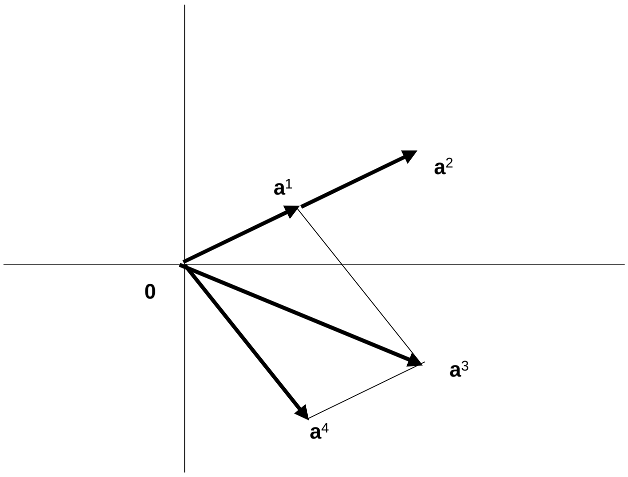

Figure 3.1: Linear Independence and Spanning in $\mathcal{R}^2$.

If any one of the first set of statements hold they all hold and none of the second set holds. Similarly if any one of the second set of statements holds all the second set and none of the first set hold. Thus any one of the first set of statements implies all the others, and any one of the second set of statements implies all the others.

I am about to explain what the term "linearly independent" means, I will then introduce spanning, and then explain rank. Determinants are defined in the chapter on determinants.

I do not prove theorem 7 here. But I do indicate in section 3.14 how a proof can be done.

3.11 Linear Independence

Definition 8 *The set of vectors* $\{\mathbf{a}^1, \mathbf{a}^2, .., \mathbf{a}^m\}$ *is* ***linearly independent*** *if the only set of numbers* $\{x_1, x_2, .., x_m\}$ *for which*

$$\mathbf{a}^1 x_1 + \mathbf{a}^2 x_2 + .. + \mathbf{a}^m x_m = 0$$

is $x_1 = x_2 = ...x_n = 0$.

You may find it helpful to look at Figure 3.1 at this point. In this Figure

- The vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ are not linearly independent because $2\mathbf{a}^1 - \mathbf{a}^2 = 0$. Geometrically $\mathbf{a}^2 = 2\mathbf{a}^1$ so both vectors lie in the same straight line.

- The three vectors $\mathbf{a^1}$, $\mathbf{a}^4$ and $\mathbf{a}^3$ are not linearly independent because $\mathbf{a^1} + \mathbf{a}^4 - \mathbf{a}^3 = 0$.

- Any two of $\mathbf{a}^1$, $\mathbf{a}^3$ and $\mathbf{a}^4$ are linearly independent.

It is in fact true, but I am not going to prove it, that no set of more than 2 vectors in $\mathcal{R}^2$ can be linearly independent, and much more generally that no set of more than n vectors in $\mathcal{R}^n$ can be linearly independent.

The idea of linear independence can be applied to the columns $\{A^1, A^2, .., A^m\}$ of the $n \times m$ matrix A. These are linearly independent if there is no set of numbers $x_1, x_2, .., x_m$ such that $A^1 x_1 + A^2 x_2 + .. + A^m x_m = 0$. In matrix notation $A^1 x_1 + A^2 x_2 + .. + A^m x_m = A\mathbf{x}$, so I can define:

Definition 9 *The columns of the matrix A are **linearly independent** if the only vector $\mathbf{x}$ for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.*

Linear independence matters for thinking about matrix inverses and solutions to simultaneous equations because of the following result:

Proposition 10 *The equation $A\mathbf{x} = \mathbf{b}$ has at most one solution if the columns of the matrix A are linearly independent. If the columns of A are not linearly independent, and the equation $A\mathbf{x} = \mathbf{b}$ has one solution, then the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.*

Proof. Suppose $\mathbf{x}_1$ and $\mathbf{x}_2$ are both solutions of the equation $A\mathbf{x} = \mathbf{b}$ so $A\mathbf{x}_1 = A\mathbf{x}_2$. Then $A\mathbf{x}_1 - A\mathbf{x}_2 = A(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0}$. If the columns of A are linearly independent the only vector for which $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$ which implies that $\mathbf{x}_1 = \mathbf{x}_2$ so there is at most one solution to $A\mathbf{x} = \mathbf{b}$.

On the other hand suppose that there is a vector $\mathbf{x}_0 \neq \mathbf{0}$ with $A\mathbf{x}_0 = \mathbf{0}$. Then if $\mathbf{x}_1$ solves the equation $A\mathbf{x} = \mathbf{b}$, for any number λ $A(\mathbf{x}_1 + \lambda\mathbf{x}_0) = A\mathbf{x}_1 + \lambda A\mathbf{x}_0 = A\mathbf{x}_1 = \mathbf{b}$ because $A\mathbf{x}_0 = \mathbf{0}$. As λ can take an infinite number of different values there are an infinite number of different solutions. ∎

3.12 Spanning

Definition 11 *The set of vectors $\{A^1, A^2, .., A^m\}$ **spans** $\mathcal{R}^n$ if for any $\mathbf{b}$ in $\mathcal{R}^n$ there are m numbers $x_1, x_2, .., x_m$ such that*

$$A^1 x_1 + A^2 x_2 + .. + A^m x_m = \mathbf{b}$$

or in matrix notation

$$A\mathbf{x} = \mathbf{b}.$$

In words this says that the columns of A span $\mathcal{R}^n$ if the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$ in $\mathcal{R}^n$, and if the columns of A do not span $\mathcal{R}^n$ there are values of $\mathbf{b}$ for which the equation $A\mathbf{x} = \mathbf{b}$ has no solution. Taken together Proposition 10 and Definition 11 establish.

Proposition 12
- *If the columns of the $n \times m$ matrix A are linearly independent and span $\mathcal{R}^n$ then for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution.*

- *If the columns of A are linearly independent but do not span $\mathcal{R}^n$, depending on* $\mathbf{b}$ *either*
  - *the equation $A\mathbf{x} = \mathbf{b}$ has one solution or*
  - *the equation $A\mathbf{x} = \mathbf{b}$ has no solution.*
- *If the columns of A are not linearly independent but do span $\mathcal{R}^n$ for every* $\mathbf{b}$ *in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions.*
- *If the columns of A are not linearly independent and do not span $\mathcal{R}^n$, depending on* $\mathbf{b}$ *either*
  - *the equation $A\mathbf{x} = \mathbf{b}$ has an infinite number of solutions or*
  - *the equation $A\mathbf{x} = \mathbf{b}$ has no solution.*

3.13 Rank

3.13.1 Definition

The column rank of the matrix A is defined as follows:

Definition 13 *The* ***column rank*** *of A is the integer r with the property that there is a set of linearly independent columns of A containing r elements, but there is no set of linearly independent columns of A containing more than r elements.*

Thus r is the largest number of linearly independent columns of A. The row rank of the matrix A is defined as follows:

Definition 14 *The* ***row rank*** *of A is the integer r with the property that there is a set of linearly independent rows of A containing r elements, but there is no set of linearly independent rows of A containing more than r elements.*

In fact row and column rank are very simply linked

Proposition 15 *The row rank and column rank of any matrix are the same.*

This would be proved in a more thorough treatment of matrices. For this reason I can drop the terms "row rank" and "column rank" and simply use the term "rank".

If A is an $n \times m$ matrix with rank r the definitions and this result imply that $r \leq m$ and $r \leq n$. The definition of column rank implies:

Proposition 16 *If A is an $n \times m$ matrix with rank r then the columns of A are linearly independent if $r = m$ and are not linearly independent if $r < m$.*

You may find it helpful to look again at Figure 3.1. Think of the vectors in the figure as the columns of a 2×2 matrix.

- No single vector spans $\mathcal{R}^2$.
- The vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ do not span $\mathcal{R}^2$.
- The vectors $\mathbf{a}^1$, $\mathbf{a}^3$ and $\mathbf{a}^4$ span $\mathcal{R}^2$, as do any two of these vectors.

The other important result about rank is:

Proposition 17 *If A is an $n \times m$ matrix with rank r then $r \leq n$, the columns of A span $\mathcal{R}^n$ if $r = n$ and do not span $\mathcal{R}^n$ if $r < n$.*

This is not obvious and I do not prove it. Thinking about this result with a two by two matrix, if the columns are $\mathbf{a}^1$ and $\mathbf{a}^2$ in Figure 3.1 these are not linearly independent, so rank $A < 2$ and the the columns of A do not span $\mathcal{R}^2$. On the other hand if the columns of A are $\mathbf{a}^1$ and $\mathbf{a}^3$ these are linearly independent and also span $\mathcal{R}^2$.

3.13.2 Finding the Rank of a Matrix

You can investigate whether the m columns of the $n \times m$ matrix A are linearly independent by looking for nonzero solutions to $A\mathbf{x} = 0$. You could equally follow the same procedure with the rows of A,thinking about $\mathbf{z}A = 0$. It is usually quicker to use rows if $n < m$ and columns if $m < n$.

When working with 2 vectors it is worth remembering that they are linearly independent unless one or both are zero or one is scalar multiple of the other. For example the vectors $(1, 2, 3)$ and $(3, 2, 1)$ are linearly independent so the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

has rank 2.

The vectors $(1, 2, 3)$ and $(4, 8, 12)$ are not linearly independent because

$$4\,(1, 2, 3) - (4, 8, 12) = 0$$

so the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$$

has rank 1.

For larger m and n there is a systematic way of finding the rank by finding something called the "row echelon" form of the matrix. Alternatively, if you have numerical values for the components of A you can ask your computer.

3.13.3 Rank and Solutions to Simultaneous Equations

Taken together Propositions 12, 16 and 17 imply:

Proposition 18 *If A is an $n \times m$ matrix, rank$A \leq n$ and rank$A \leq m$.*

- *If rank $A = m = n$ for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution.*

- *If rank $A = m < n$ depending on* $\mathbf{b}$ *either*
  - *the equation* $A\mathbf{x} = \mathbf{b}$ *has one solution or*
  - *the equation* $A\mathbf{x} = \mathbf{b}$ *has no solution.*
- *If the rank $A = n < m$ for every* $\mathbf{b}$ *in* $\mathcal{R}^n$ *the equation* $A\mathbf{x} = \mathbf{b}$ *has an infinite number of solutions.*
- *If rank$A < m$ and rank$A < n$ depending on* $\mathbf{b}$ *either*
  - *the equation* $A\mathbf{x} = \mathbf{b}$ *has an infinite number of solutions or*
  - *the equation* $A\mathbf{x} = \mathbf{b}$ *has no solution.*

3.13.4 Rank and Inverse

I have already argued that if A has an the inverse is unique, so for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution $\mathbf{x} = A^{-1}\mathbf{b}$. It is also true (although I have not proved it) that the matrix A has an inverse if for every $\mathbf{b}$ in $\mathcal{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has exactly one solution. From the first part of proposition 18 this is only possible if A is a square $n \times n$ matrix. Taking into account the last part of proposition 18 gives the result on the matrix inverse

Proposition 19 *The square matrix $n \times n$ matrix A has an inverse if and only if rank $A = n$, a condition that can also be stated as*

- *the columns of A are linearly independent, or equivalently*
- *the only vector for which* $A\mathbf{x} = \mathbf{0}$ *is* $\mathbf{x} = \mathbf{0}$.

These conditions are satisfied if and only if the equation $A\mathbf{x} = \mathbf{b}$ *has a unique solution for every* $\mathbf{b}$ *in* $\mathcal{R}^n$.

3.14 A more sophisticated approach to matrices

The treatment in this chapater is very cookbook - intuition, not proof. It's intention is to give you quickly the essentials of what you need to know on matrices. But the rules for matrices, in particular matrix multiplication, can seem very arbitrary. This section sketches where they come from and how they relate to the existence of an inverse. This tells you what the steps in the proof are. You don't need to read it, but if you have a good mathematical background and are curious about how the argument works this indicates how it can be done.

- A function a from $\mathcal{R}^m$ into $\mathcal{R}^n$ is linear if for any $\mathbf{x}_1$ and $\mathbf{x}_2$ in and any real numbers γ_1 and γ_2

$$a(\gamma_1\mathbf{x}_1 + \gamma_2\mathbf{x}_2) = \gamma_1 a(\mathbf{x}_2) + \gamma_2 a(\mathbf{x}_2).$$

- A necessary and sufficient condition for $a : \mathcal{R}^m \to \mathcal{R}^n$ to be linear is that there is an $n \times m$ matrix such that
$$a(\mathbf{x}) = A\mathbf{x}$$
for all $\mathbf{x}$ in $\mathcal{R}^m$.

- Matrix multiplication comes from composition of functions. If $b : \mathcal{R}^n \to \mathcal{R}^p$ then
$$b(a(\mathbf{x})) = BA\mathbf{x}.$$

- If the function $a : \mathcal{R}^m \to \mathcal{R}^n$ has an inverse $a^{-1} : \mathcal{R}^n \to \mathcal{R}^m$ then a^{-1} is linear. The associated matrix is A^{-1}. If a does not have an inverse then A^{-1} does not exist.

- The function $a : \mathcal{R}^m \to \mathcal{R}^n$ has an inverse if it is one to one and onto. The function is onto if the columns of A span $\mathcal{R}^n$. It is one to one if the columns of A are linearly independent.

- If S_1 and S_2 are sets of vectors in $\mathcal{R}^n$ with p_1 and p_2 elements, the elements of S_1 are linearly independent, and the columns of S_2 span $\mathcal{R}^n$ then $p_1 \leq p_2$. As the columns of the $n \times n$ identity matrix are linearly independent and span this implies that
$$p_1 \leq n \leq p_2.$$
If $p_1 < n$ then the p_1 components of S_1 do not span. If $n < p_2$ the components of S_2 are not linearly independent. The members of a set of n elements of $\mathcal{R}^n$ is either linearly independent and spans or is not linearly independent and does not span.

- The dimension of a space is the minimum number of elements needed to span it.

- The rank of a matrix is the dimension of the space spanned by its columns.

Chapter 4

Determinants

4.1 Why Determinants?

The previous chapter on matrices ended up with a set of conditions for a matrix to have an inverse. The easiest to write down is the condition that the $n \times n$ matrix A has an inverse if and only if there is no n vector for which $A\mathbf{x} = \mathbf{0}$ apart from $\mathbf{x} = \mathbf{0}$. This is fine as far as it goes, but not much practical use without guidance on finding out whether the condition is satisfied. So it would be nice to have a formula involving the components of the matrix A that told you directly whether A has an inverse. There is such a formula; it is called the determinant of A, written as $\det A$ or $|A|$, and the square matrix A has an inverse if and only if $\det A \neq 0$. This is the first part of Theorem 20 in this chapter. The determinant also fills another nasty gap. It would be very nice to have a formula for the inverse of A. There is one; it involves determinants. Theorem 20 in this chapter gives the formula. Determinants have further uses. I will be introducing the ideas of the eigenvalues and eigenvectors in chapter 5 which follows this one.. Determinants are used to find the eigenvalues of a matrix. You will meet positive and negative definite matrices. How do you find out if a matrix is positive or negative definite? Using the determinant of course.

This all sounds very good. The difficulty is that the formula for a determinant does not look at all nice when you come to write it down. In fact we never do write it down in full for anything bigger than a 3×3 matrix. We have a simple formula for the determinant of a 2×2 matrix, but the general formula is inductive, giving the determinant of an $n \times n$ matrix as a function of the determinants of n different $(n-1) \times (n-1)$ matrices. There some matrices for which the calculation of the determinant is straightforward, I demonstrate some of them. But in general calculating a determinant by hand is a very daunting task. Nevertheless, owing to the close links between determinants and solutions of simultaneous equations, a considerable amount of work went into the theory of determinants in the nineteenth century, and some elegant results that are very far from obvious were obtained. In the twentieth century interest moved from determinants to matrices partly due to swings of fashion in pure mathematics; matrices lend themselves much more easily to abstraction than determinants. More importantly from the point of view of applied mathematicians such as economists computers took over the nitty gritty job of calculating determinants and inverting matrices, so it became less important to know a lot about determinants, apart from the fact that they exist, and that tell you whether a matrix has an inverse. This chapter summarizes the facts about determinants that I think you should be aware of, but does not provide any proofs.

4.2 Notation and Definition

Only square matrices have determinants. The notation for the determinant of an $n \times n$ matrix

$$A = \begin{pmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{pmatrix}$$

is

$$\det A = |A| = \begin{vmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{vmatrix}.$$

The difference between notation for a determinant and notation for a matrix is that the determinant has lines $|.|$ where the matrix has brackets $(.)$.

If A is a 1×1 matrix (A_{11})

$$\det A = A_{11}.$$

If A is a 2×2 matrix

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

$$\det A = \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} = A_{11}A_{22} - A_{12}A_{21}.$$

You may find it helpful to remember this as the difference between the product of the diagonal terms and the product of the off diagonal terms.

The determinant of a 3×3 matrix A is

$$\det A = \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix} \tag{4.1}$$

The determinant of an $n \times n$ matrix

$$A = \begin{pmatrix} A_{11} & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} \end{pmatrix}$$

is

$$\begin{aligned} \det A &= A_{11}M_{11} - A_{12}M_{12} + A_{13}M_{13} - \ldots\ldots + (-1)^{1+c} M_{1c} \ldots. + (-1)^{1+n} \qquad (4.2) \\ &= \sum_{c=1}^{n} (-1)^{1+c} A_{1c}M_{1c} \end{aligned}$$

where M_{1c} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column c from the matrix A.

$$M_{1c} = \begin{vmatrix} A_{21} & . & A_{2(c-1)} & A_{2(c+1)} & . & A_{2n} \\ A_{31} & . & A_{3(c-1)} & A_{3(c+1)} & . & A_{3n} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ A_{n1} & . & A_{n(c-1)} & A_{n(c+1)} & . & A_{nn} \end{vmatrix}$$

The $1c$ **cofactor** C_{rc} of the matrix A is

$$C_{1c} = (-1)^{1+c} M_{1c}.$$

Thus equation 4.2 can also be written as

$$\begin{aligned} \det A &= A_{11}C_{11} + A_{12}C_{22} + A_{13}C_{13} + \ldots\ldots. + A_{1n-1}C_{1n-1} + A_{1n}C_{1n} \\ &= \sum_{c=1}^{n} A_{1c}C_{1c}. \end{aligned}$$

Note that only square matrices have determinants, so you should assume that all the matrices that you meet in this chapter are square.

4.3 Simultaneous Equations and Determinants

4.3.1 The 2×2 case

Determinants can appear mysterious; they turn out to have many mathematical properties that are far from obvious, but why should anyone have thought them interesting in the first place? The reason is that determinants emerge very naturally from the search for solutions to linear simultaneous equations.

Consider the equations

$$A_{11}x_1 + A_{12}x_2 = b_1 \tag{4.3}$$

$$A_{21}x_1 + A_{22}x_2 = b_2. \tag{4.4}$$

You will have learnt in high school mathematics that you can solve simultaneous equations either by substitution or by elimination of variables. The first step in substitution uses equation 4.4 to get x_2 as a function of x_1

$$x_2 = \frac{b_2 - A_{21}x_1}{A_{22}}. \tag{4.5}$$

You then substitute this expression in equation 4.3 to get

$$A_{11}x_1 + A_{12}\left(\frac{b_2 - A_{21}x_1}{A_{22}}\right) = b_1$$

and rearrange to get

$$x_1 = \frac{b_1 - \left(\frac{A_{12}}{A_{22}}\right) b_2}{\left(A_{11} - \frac{A_{12}A_{21}}{A_{22}}\right)} = \frac{A_{22}b_1 - A_{12}b_2}{A_{11}A_{22} - A_{12}A_{21}}.$$

You can then substitute this expression for x_1 in equation 4.5 to get an expression

$$x_2 = \left(\frac{1}{A_{22}}\right)(b_2 - A_{21}x_1) = \frac{1}{A_{22}}\left(b_2 - A_{21}\left(\frac{A_{22}b_1 - A_{12}b_2}{A_{11}A_{22} - A_{12}A_{21}}\right)\right)$$

that after a bit of algebra reduces to

$$x_2 = \frac{b_2A_{11} - A_{21}b_1}{A_{11}A_{22} - A_{12}A_{21}}.$$

However the algebra is not much fun, and more seriously if either or both of A_{22} and $A_{11}A_{22} - A_{12}A_{21}$ are zero substitution involves the forbidden act of dividing by zero. If $A_{22} = 0$ equation 4.4 reduces to $A_{21}x_1 = b_2$ which provided $A_{21} \neq 0$ you can solve for $x_1 = b_2/A_{21}$. However if $A_{11}A_{22} - A_{12}A_{21} = 0$ trying to solve the equations by substitution runs into real difficulties. So if you are after a general solution it is better to work by elimination of variables, which avoids doing any division until the very last step.

The first step in solving the equations 4.3 and 4.4 by elimination of variables is to multiply equation 4.3 by A_{22} and equation 4.4 by A_{12} to get

$$\begin{aligned} A_{11}A_{22}x_1 + A_{12}A_{22}x_2 &= A_{22}b_1 \\ A_{12}A_{21}x_1 + A_{12}A_{22}x_2 &= A_{12}b_2. \end{aligned}$$

Subtracting the second of these equations from the first gives

$$(A_{11}A_{22} - A_{12}A_{21})\, x_1 = (A_{22}b_1 - A_{12}b_2) \tag{4.6}$$

Similarly multiplying equation 4.3 by A_{21} and equation 4.4 by A_{11} gives

$$\begin{aligned} A_{21}A_{11}x_1 + A_{12}A_{21}x_2 &= A_{21}b_1 \\ A_{21}A_{11}x_1 + A_{11}A_{22}x_2 &= A_{11}b_2. \end{aligned}$$

Subtracting the first of these equations from the second gives

$$(A_{11}A_{22} - A_{12}A_{21})\, x_2 = A_{11}b_2 - A_{21}b_1. \tag{4.7}$$

Writing equations 4.6 and 4.7 in determinant notation gives the result

$$(\det A)\, x_1 = \begin{vmatrix} b_1 & A_{12} \\ b_2 & A_{22} \end{vmatrix}. \tag{4.8}$$

$$(\det A)\, x_2 = \begin{vmatrix} A_{11} & b_1 \\ A_{21} & b_2 \end{vmatrix}. \tag{4.9}$$

This tells you two things. Firstly if $\det A = 0$ the only values of b_1 and b_2 for which there is a solution are those for which the right hand side of equations

4.8 and 4.9 are zero. Secondly if $\det A \neq 0$ you have a formula for the solution. Remember that

$$\det A = \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}$$

so the formula is

$$\begin{aligned} x_1 &= \frac{\begin{vmatrix} b_1 & A_{12} \\ b_2 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ x_2 &= \frac{\begin{vmatrix} A_{11} & b_1 \\ A_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}}. \end{aligned}$$

The first column of the matrix inverse is the solutions to the equations

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 &= b_1 \\ A_{21}x_1 + A_{22}x_2 &= b_2. \end{aligned}$$

with $b_1 = 1$ and $b_2 = 0$. The second column of the inverse is the solution to the equations with $b_1 = 0$, and $b_2 = 1$. This gives the components of A^{-1} as

$$\begin{aligned} A^{-1}_{11} &= \frac{\begin{vmatrix} 1 & A_{12} \\ 0 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{A_{22}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{21} &= \frac{\begin{vmatrix} A_{11} & 1 \\ A_{21} & 0 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{-A_{21}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{12} &= \frac{\begin{vmatrix} 0 & A_{12} \\ 1 & A_{22} \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{-A_{12}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} \\ A^{-1}_{22} &= \frac{\begin{vmatrix} A_{11} & 0 \\ A_{21} & 1 \end{vmatrix}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}} = \frac{A_{11}}{\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}}. \end{aligned}$$

I have to admit that I find these formulae deeply unmemorable, my memory works better with words:

- The determinant of a 2×2 matrix is the product of its diagonal terms minus the product of the off diagonal terms.
- The inverse of a 2×2 matrix is found by swapping the diagonal terms, changing the sign of the off diagonal terms and the dividing by the determinant.

4.4 Determinants, Inverses and Cramer's Rule

These results generalize from the 2×2 to the $n \times n$ case. The most important result on determinant is:

Theorem 20 (Determinants and Matrix Inverses) *The $n \times n$ matrix has an inverse if and only if $\det A \neq 0$. If $\det A \neq 0$ the the inverse of A is*

$$A^{-1} = \left(\frac{1}{\det A}\right) \operatorname{adj} A$$

*where $\operatorname{adj} A$ is the **adjoint** of A, that is the matrix whose ij entry is $(-1)^{i+j} M_{ji}$ where M_{ji} is the ji minor, that is the determinant of the matrix formed by deleting row j and column i from the matrix A.*

Note in the last part of this proposition that the ij component of $\operatorname{adj} A$, that is the entry in **row i** and **column j** of $\operatorname{adj} A$, is the determinant of the matrix formed by deleting **row j** and **column i** from A. This is not a typo.

This result makes it possible to solve the equation $Ax = b$ when $\det A \neq 0$ by using the formula in Theorem 20 to find A^{-1} and thus find the solution $x = A^{-1}b$. There is an alternative approach, **Cramer's Rule**, that makes direct use of determinants:

Theorem 21 (Cramer's Rule) *If $\det A \neq 0$ the equation $Ax = b$ has a unique solution for every b. The solution is*

$$x_i = \frac{\det B_i}{\det A} \qquad i = 1, 2....n$$

where B_i is the $n \times n$ matrix obtained by replacing the i^{th} column of A by the vector b.

Cramer's rule is used in economics, however if n is large calculating the determinants is burdensome, and if n is small it is often easier to solve the equations directly by substitution or elimination of variables, or equivalently using row echelon matrices.

4.5 Calculating Determinants: Some Special Cases

4.5.1 The 3×3 Case

It is possible in principle to calculate the determinant of any matrix working with the definition. This is easy for a 2×2 matrix, and not bad for a 3×3 matrix where the definition implies that

$$\begin{aligned}
&\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} \\
= \; & A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix} \\
= \; & A_{11}A_{22}A_{33} + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} \\
& -A_{11}A_{23}A_{32} - A_{12}A_{21}A_{33} - A_{13}A_{22}A_{31}.
\end{aligned}$$

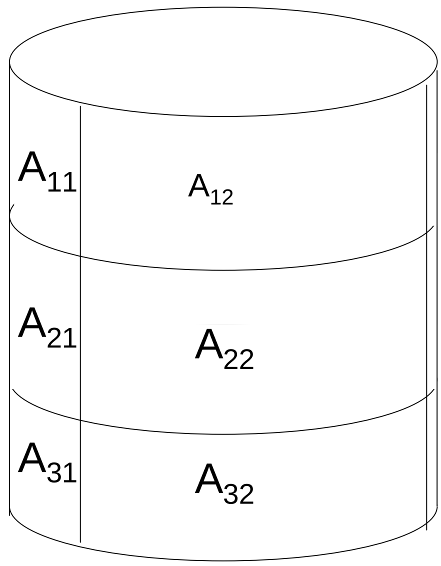

Figure 4.1:

I think of this formula by visualizing the matrix pasted on a cylinder as in Figure 4.1. The terms with a plus sign + come from taking an element on the top row of the cylinder and then finding two more elements by moving one place to the right and one place down twice. The terms with a negative sign − come from taking an element on the top row and then finding two more elements by moving one place to the left and one place down twice. Unfortunately this visualization trick only works for 2×2 and 3×3 matrices.

4.5.2 Diagonal Matrices

Definition 22 *The $n \times n$ matrix A is **diagonal** if all the off diagonal terms are zero, that is $A_{ij} = 0$ when $i \neq j$.*

For example the matrix

$$\begin{pmatrix} A_{11} & 0 & 0 \\ 0 & A_{22} & 0 \\ 0 & 0 & A_{33} \end{pmatrix}$$

is diagonal. There is an easy result on the formula for the determinant of a diagonal matrix.

Theorem 23 (Determinants of Diagonal Matrices) *The determinant of a diagonal matrix is the product of its diagonal terms.*

To see why this is so let

$$A = \begin{pmatrix} A_{11} & 0 & . & . & \\ 0 & A_{22} & 0 & . & \\ . & 0 & A_{33} & 0 & \\ . & . & .0 & . & 0 \\ . & . & . & 0 & A_{nn} \end{pmatrix}.$$

Then

$$\det A = A_{11} \begin{vmatrix} A_{22} & 0 & . & \\ 0 & A_{33} & 0 & \\ . & .0 & . & 0 \\ . & . & 0 & A_{nn} \end{vmatrix} = A_{11}A_{22} \begin{vmatrix} A_{33} & . & . \\ . & . & . \\ . & . & A_{nn} \end{vmatrix} = A_{11}A_{22}.....A_{nn}.$$

4.5.3 The Identity Matrix

The identity matrix I is a diagonal matrix for which all the diagonal terms are 1. The result on diagonal matrices implies at once:

Proposition 24 *The determinant of the identity matrix is 1.*

4.6 The General Case

4.6.1 Use a Computer

Calculating the determinant of a general $n \times n$ matrix with $n > 3$ by hand is usually a deeply tedious piece of algebra or arithmetic with a huge risk of making mistakes. This is a job for a computer, as is finding the inverse of a matrix. Mine has just told me in seconds that the matrix

$$\begin{pmatrix} -5 & 3 & -2 & 4 \\ -2 & 1 & -1 & 2 \\ 4 & -2 & 4 & 6 \\ -7 & 4 & -3 & 11 \end{pmatrix}$$

has determinant 10 and inverse

$$\begin{pmatrix} -0.4 & -5.4 & -0.5 & 1.4 \\ 0.6 & -7.4 & -0.5 & 1.4 \\ 1.0 & 2.0 & 0.5 & -1.0 \\ -0.2 & -0.2 & 0.0 & 0.2 \end{pmatrix}.$$

4.7 Expanding Along Rows and Columns: optional

Look quickly at this material

4.7.1 The Expansion Result

If you do have to calculate a determinant by hand there are some results that may make the calculation easier which derive from the result on expanding along rows and columns which I am about to state.

Equation 4.2 gives the general formula for the determinant of an $n \times n$ matrix

$$\begin{aligned} \det A &= A_{11}M_{11} - A_{12}M_{12} + A_{13}M_{13} - \ldots\ldots \\ &\quad + (-1)^{1+c} A_{1c}M_{1c} \ldots. + (-1)^{1+n} A_{1n}M \\ &= \sum_{c=1}^{n} (-1)^{1+c} A_{1c}M_{1c} \end{aligned}$$

where M_{1c} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row 1 and column c from the matrix A. M_{rc} is called the $1c$ **minor** of the matrix A. I think of this as expanding along row 1, and remember the formula as the sum over $c = 1, 2 \ldots n$ of terms of the form

element c of row 1
$\times (-1)^{1+c}$
$\times$determinant of the matrix formed from A by deleting row 1 and column c.

It is in fact possible to expand along any row or column. Stating this result uses the general definition of a minor.

Definition 25 *The rc minor M_{rc} of the matrix A is the determinant of the $(n-1)\times(n-1)$ matrix obtained by deleting row r and column c from the matrix A.*

The formal result on expanding along a row or column is:

Theorem 26 (Expansion of a Determinant Along a Row or Column) *If A is an $n \times n$ matrix then for any row r*

$$\det A = \sum_{c=1}^{n} (-1)^{r+c} A_{rc} M_{rc} \tag{4.10}$$

and for any column c

$$\det A = \sum_{r=1}^{n} (-1)^{r+c} A_{rc} M_{rc}. \tag{4.11}$$

It is straightforward to verify the result for a 2×2 matrix where $M_{11} = A_{22}$, $M_{12} = A_{21}$, $M_{21} = A_{12}$ and $M_{22} = A_{11}$.

I think of equation 4.10 as expanding along row r, remembering that $\det A$ is the sum over $c = 1, 2...n$ of terms

element c of row r
$\times (-1)^{r+c}$
$\times$determinant of the matrix formed from A by deleting row r and column c.

I think of equation 4.11 as expanding along column c, remembering that $\det A$ is the sum over $r = 1, 2...n$ of terms

element r of column c
$\times (-1)^{r+c}$
$\times$determinant of the matrix formed from A by deleting row r and column c.

An example using this result is:

Example 27 *Expanding along column 2 of the matrix*

$$\begin{pmatrix} A_{11} & b_1 & A_{13} \\ A_{21} & b_2 & A_{23} \\ A_{31} & b_3 & A_{33} \end{pmatrix}$$

gives

$$\begin{vmatrix} A_{11} & b_1 & A_{13} \\ A_{21} & b_2 & A_{23} \\ A_{31} & b_3 & A_{33} \end{vmatrix} = -b_1 \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + b_2 \begin{vmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{vmatrix} - b_3 \begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix}.$$

Proposition 26 is particularly useful if you are working with a matrix that has several zeros in the same row or column. For example, expanding along the bottom row

$$\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ 0 & A_{32} & 0 \end{vmatrix} = -A_{32} \begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix} = -A_{32} \left(A_{11}A_{23} - A_{13}A_{21} \right).$$

4.7.2 Consequences of the Expansion Result

The Determinant of a Matrix with a Row or Column of Zeros is Zero

This is a list of results that are sometimes useful and follow easily from the expansion result Theorem 26. The first set of results can all be proved very simply by expanding along the relevant row or column. The first of these is particularly worth remembering.

Proposition 28
- *If all the elements of a row of a square matrix* A *are* 0 *then* $\det A = 0$.
- *If all the elements of a column of a square matrix* A *are* 0 *then* $\det A = 0$.

Determinants of Triangular Matrices are the Product of the Diagonal Terms

Two more results that follow quite straight forwardly from the expansion result; these involve upper and lower triangular matrices.

Definition 29 *The* $n \times n$ *matrix* A *is* ***upper triangular*** *if all the terms below the diagonal are zero, that is* $A_{ij} = 0$ *when* $i > j$.

For example the matrix

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ 0 & A_{22} & A_{23} \\ 0 & 0 & A_{33} \end{pmatrix}$$

is upper triangular.

Proposition 30 *The determinant of an upper triangular matrix is the product of its diagonal terms.*

The result follow quite easily by expanding along column 1 the determinant of an $n \times n$ upper triangular matrix

$$\begin{aligned} \begin{vmatrix} A_{11} & A_{12} & A_{13} & . & . & A_{1n} \\ 0 & A_{22} & A_{23} & . & . & A_{2n} \\ 0 & 0 & A_{33} & . & . & . \\ 0 & 0 & 0 & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & 0 & 0. & 0 & A_{nn} \end{vmatrix} &= A_{11} \begin{vmatrix} A_{22} & A_{22} & . & . & A_{2n} \\ 0 & A_{33} & . & . & A_{3n}. \\ 0 & 0 & . & . & . \\ . & . & . & . & . \\ 0 & 0 & 0. & 0 & A_{nn} \end{vmatrix} \\ &= A_{11}A_{22} \begin{vmatrix} A_{33} & . & . & A_{3n}. \\ 0 & . & . & . \\ . & . & . & . \\ 0 & 0. & 0 & A_{nn} \end{vmatrix} \\ &= A_{11}A_{22}A_{33}......A_{nn}. \end{aligned}$$

Definition 31 *The* $n \times n$ *matrix* A *is* ***lower triangular*** *if all the terms above the diagonal are zero, that is* $A_{ij} = 0$ *when* $i < j$.

For example the matrix

$$\begin{pmatrix} A_{11} & 0 & 0 \\ A_{21} & A_{22} & 0 \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

is lower triangular.

Proposition 32 *The determinant of a lower triangular matrix is the product of its diagonal terms.*

The proof is very similar to the corresponding proposition for an upper triangular matrix, except the determinant is expanded along the first row.

4.8 Determinants of Related Matrices

The most important result here is

Theorem 33 *Determinants and Products. The determinant of the product of two $n \times n$ matrices A and B is the product of the determinants, that is*

$$|AB| = |A|\,|B|\,.$$

As $\det I = 1$ this implies that if A has an inverse A^{-1} then $(\det A)\left(\det A^{-1}\right) = \det\left(AA^{-1}\right) = \det I = 1$. If $\det A = 0$ this implies the impossible $0 = 1$, and thus proves:

Theorem 34 *If A has an inverse then $\det A \neq 0$ and*

$$\det\left(A^{-1}\right) = \frac{1}{\det A}.$$

There is also a useful result on the transpose A' of the matrix A. (The transpose A' of A is the matrix in whose row vectors are the column vectors of A, so $A'_{ij} = A_{ji}$). The result is

Proposition 35 *The determinant of a matrix A and the determinant of its transpose A' are equal.*

$$\det A = \det\left(A'\right).$$

Chapter 5

Eigenvalues and Eigenvectors

5.1 Introduction

Eigenvalues and eigenvectors are simply defined. A vector $\mathbf{x}$ is an eigenvector of matrix A with eigenvalue λ if $\mathbf{x} \neq \mathbf{0}$ and

$$A\mathbf{x} = \lambda\mathbf{x}. \tag{5.1}$$

This says that multiplying the eigenvector $\mathbf{x}$ by the matrix A has the same effect as multiplying the vector by the number λ, geometrically this makes the vector longer or shorter, and if $\lambda < 0$ makes the vector point in exactly the opposite direction, but cannot otherwise change the direction of the vector. Note that if A is an $n \times m$ matrix and $\mathbf{x}$ is an m vector then $A\mathbf{x}$ is an n vector and $\lambda\mathbf{x}$ is an m vector, so the only matrices that can have eigenvectors and eigenvalues are square $n \times n$ matrices.

It is not immediately obvious why economists might be interested in eigenvalues and eigenvectors. In fact they turn out to be very important in several contexts. One is dealing with quadratic forms which are what quadratic functions turn into when they grow up. You will meet these in mathematics for microeconomics. They also matter in econometrics and finance (they underpin the Capital Asset Pricing Model). The other area where eigenvalues and eigenvectors are important is economic dynamics, studied with difference and differential equations. This is currently extremely trendy. Hence the need to know something about eigenvalues and eigenvectors.

5.2 Finding Eigenvalues: the Characteristic Polynomial

Equation 5.1 can be rewritten as

$$(A - \Lambda)\,\mathbf{x} = \mathbf{0} \tag{5.2}$$

where Λ is a diagonal matrix, that is all its off diagonal terms are 0, and all its diagonal terms are λ. As $\mathbf{x} \neq 0$ equation 5.2 and the result that $\det B = 0$ if and only if there is a non zero vector $\mathbf{x}$ such that $B\mathbf{x} = \mathbf{0}$ implies that

$$\det(A - \Lambda) = 0$$

or writing this out in full

$$\begin{vmatrix} A_{11}-\lambda & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22}-\lambda & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn}-\lambda \end{vmatrix} = 0. \tag{5.3}$$

Thus all you have to do to find the eigenvalues of the matrix is solve this equation. Sadly that is often not an easy task. Firstly you have to calculate the determinant; this is a polynomial of degree n called the **characteristic polynomial** $\det(A - \Lambda)$. Then you have to solve the **characteristic equation** $\det(A - \Lambda) = 0$. The fundamental theorem of algebra (see Background Notes on polynomials) tells you that for any polynomial $p(\lambda)$ in λ of degree n there are n numbers, called the roots of the polynomial with the property that

$$p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\ldots.(\lambda_n - \lambda). \tag{5.4}$$

The number λ_i are called the roots of the polynomial; they are the solutions of the equation $p(\lambda) = 0$, and the eigenvalues of the matrix A. One fact that follows straight from the fact that the determinant in equation 7.11 is equal to the polynomial in equation 5.4 and thinking about what they are when $\lambda = 0$, gives

Proposition 36 *If A is a square matrix the determinant of A is the product of the eigenvalues of A, that is*

$$\det A = \lambda_1\lambda_2\ldots.\lambda_n.$$

5.3 Some Real Matrices Have Complex Eigenvalues and Eigenvectors

Even if the matrix A is formed from real numbers the eigenvalues can be complex. For example the matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has a characteristic equation $\lambda^2 + 1 = 0$ so has no real roots. In fact if x_1 and x_2 are real numbers

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}$$

and it is easy to check that the two vectors

$$(x_1, x_2)\begin{pmatrix} x_2 \\ -x_1 \end{pmatrix} = 0$$

so the vectors $\mathbf{x}$ and $A\mathbf{x}$ area orthogonal. (at $90°$) to each other, so there is no way that $A\mathbf{x}$ can be a scalar multiple $\lambda\mathbf{x}$ of $\mathbf{x}$ if $\mathbf{x}$ is real. The one general thing that can be said about the eigenvalues of real matrices is that the fundamental theorem of algebra (Background Notes on polynomials) implies that any complex eigenvalues come in pairs of complex conjugates.

5.4 When Finding the Eigenvalues of a Matrix is Easy

5.4.1 The 2×2 Case

In this case the characteristic equation of the matrix is

$$\begin{aligned} \begin{vmatrix} A_{11}-\lambda & A_{12} \\ A_{21} & A_{22}-\lambda \end{vmatrix} &= \lambda^2 - \lambda\left(A_{11}+A_{22}\right) + A_{11}A_{22} - A_{12}A_{21} \\ &= \lambda^2 - \lambda\left(trA\right) + \det A \end{aligned}$$

where trA is the trace of A, that is the sum of the diagonal terms, and $\det A$ is the determinant of A. There is of course a formula for the solutions of this equation. For example if

$$A = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$$

The characteristic equation is

$$\begin{vmatrix} 4-\lambda & 1 \\ -2 & 1-\lambda \end{vmatrix} = \lambda^2 - \lambda\left(trA\right) + \det A = \lambda^2 - 5\lambda + 6 = \left(\lambda-2\right)\left(\lambda-3\right)$$

so the eigenvalues are 2 and 3. The eigenvector corresponding to the eigenvalue 2 can be found by solving the equation

$$\begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

or equivalently

$$\begin{pmatrix} 4-2 & 1 \\ -2 & 1-2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1+x_2 \\ -2x_1-x_2 \end{pmatrix} = 0$$

so any x_1 and x_2 with $x_2 = -2x_1$ is a solution; $x_1 = 1$, $x_2 = -2$, is the simplest but any vector of the form

$$\beta\begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

where $\beta \neq 0$ is an eigenvector. This is a general point about eigenvectors, if $\mathbf{x}$ is an eigenvector of matrix A so is $\beta\mathbf{x}$.

A similar argument for the eigenvector corresponding to the eigenvalue 3, says that any vector of the form

$$\beta\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

is an eigenvector.

5.4.2 Diagonal, Upper and Lower Triangular Matrices

I showed in chapter 4 on determinants the determinant of a diagonal or upper or lower triangular matrix is the product of the diagonal terms. If A is a diagonal or upper or lower triangular matrix so also is $A-\Lambda$, so the characteristic equation of A

$$(A_{11}-\lambda)(A_{22}-\lambda)(A_{33}-\lambda)\ldots.(A_{nn}-\lambda)=0 \tag{5.5}$$

so the solutions are the diagonal terms A_{11}, A_{22}, $A_{33}....A_{nn}$. In these cases the eigenvalues are the diagonal terms of the matrix.

5.5 Diagonalizing a Matrix

There are some useful results that I state without proof.

Proposition 37 *If all the eigenvalues of A are all different then there is a matrix M with an inverse M^{-1} with the property that*

$$A = M^{-1}DM$$

where D is a diagonal matrix, whose diagonal terms are the eigenvalues of A.

There are stronger statements to be made about the eigenvalues and eigenvectors of symmetric matrices (the matrix A is symmetric if $A' = A$, so $A_{ij} = A_{ji}$ for all i and j.

Proposition 38 *If the matrix A is symmetric, and all its components are real numbers, then all its eigenvalues are real and the eigenvectors corresponding to different eigenvalues are orthogonal., that is if $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors of A with eigenvalues λ_1 and λ_2, and $\lambda_1 \neq \lambda_2$ then*

$$\mathbf{x}_1'\mathbf{x}_2 = 0.$$

There is then a fairly natural definition

Definition 39 *The matrix M is* ***orthogonal.*** *if its transpose is also its inverse, so*

$$MM' = M'M = I.$$

Another way of saying this is that any two different columns of M are orthogonal. The big result is then

Proposition 40 *If the matrix A is symmetric and all its components are real numbers, then all its eigenvalues are real and there is an orthogonal. matrix M such that*

$$A = M'DM$$

where D is a diagonal matrix whose diagonal terms are the eigenvalues of I.

For some purposes, notably the solution of difference and differential equations we want to know about A^n. If A is diagonalizable, that is it can be written in the form

$$A = M^{-1}DM$$

where D is a diagonal matrix

$$A^n = M^{-1}D^nM$$

where as D is a diagonal matrix D^n is also diagonal and component ii of D^n is D_{ii}^n. This is nice because if $|D_{ii}| < 1$ for all i then D_{ii}^n tends to 0 as n tends to infinity, so A^n also tends to 0.

Chapter 6

Introduction to Multivariate Calculus

6.1 Why Economists are Interested

We use multivariate calculus in economics because the things we are interested in are usually functions of more than one variable. This gets suppressed in introductory economics. You may have worked with a model where the cost to a firm of producing y is $c(y) = \frac{1}{2}y^2$ and written $\frac{dc}{dy} = y$ as marginal cost. This seems a bit special, so perhaps you generalized this to work with a cost function $c(y) = \gamma y^2$. At this stage you are treating γ as a parameter of the cost function. However when you think about it the cost must depend on input prices and the technology used, so γ is itself a function, and changes in input prices change γ. You will learn to think of cost as a function of input prices $w_1,\ w_2$ and output $c(w_1, w_2, y)$. For example the minimum cost of producing output y from inputs x_1 and x_2 with prices w_1 and w_2 using the technology given by the Cobb-Douglas production function $y = Ax_1^a x_2^b$ is

$$c(w_1, w_1, y) = A^{\frac{-1}{a+b}} \left[\left(\frac{a}{b}\right)^{\frac{b}{a+b}} + \left(\frac{a}{b}\right)^{\frac{-a}{a+b}} \right] w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}}.$$

(You can find this result on page 54 of H.R.Varian, Microeconomic Analysis.) This reduces to the cost function γy^2 if $a = b = \frac{1}{4}$ and $\gamma = 2A^{-2}w_1^{\frac{1}{2}} w_2^{\frac{1}{2}}$. However, having acknowledged that costs depend on w_1, w_2 and y, you now have to use the partial derivative

$$\frac{\partial c(w_1, w_1, y)}{\partial y} = \frac{1}{a+b} A^{\frac{-1}{a+b}} \left[\left(\frac{a}{b}\right)^{\frac{b}{a+b}} + \left(\frac{a}{b}\right)^{\frac{-a}{a+b}} \right] w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}-1}$$

for marginal cost. When you have a function of many variables $f(x_1, x_2, .x_i..x_n)$ the partial derivative is defined analogously to the derivative of a function of a single variable as

$$\frac{\partial f(x_1, x_2, .x_i..x_n)}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, .x_i + h..x_n) - f(x_1, x_2, .x_i..x_n)}{h}.$$

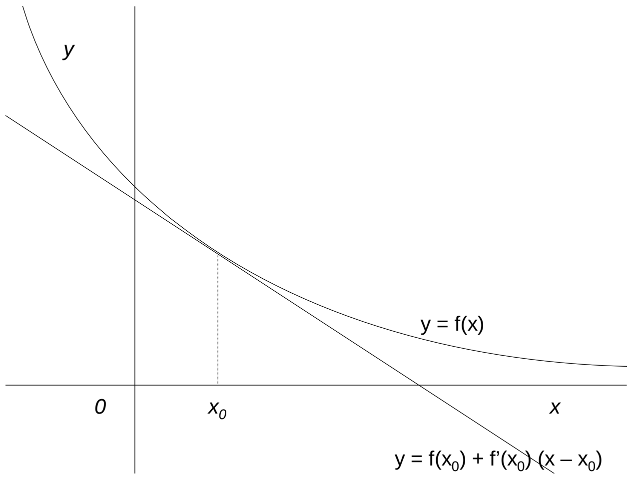

Figure 6.1: A function and its tangent

You calculate the partial derivative $\dfrac{\partial f(x_1, x_2, .x_i..x_n)}{\partial x_i}$ in exactly the same way as you calculate the derivative of a function of a single variable, treating all the variables except x_i as parameters of the function.

In fact the distinction between a parameter and a variable depends entirely on what you are interested in. Varian writes $c(w_1, w_1, y)$ treating A, a and b as parameters. But for some purposes, for example modelling technical change you might want to think of A, a and b as variables, and write costs as $g(A, a, b, w_1, w_2, y)$.

6.2 Derivatives and Approximations

6.2.1 When Can You Approximate

Think about a function of one variable $y = f(x)$ illustrated by the curved line in Figure 6.1 and its tangent at x_0 which is a straight line with equation $y = f(x_0) + f'(x_0)(x - x_0)$. The definition of the derivative $f'(x)$ implies that close to x the tangent line is a good approximation to the original function, that is

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) \text{ when } x \text{ is close to } x_0.$$

Now think about the function of two variables $y = f(x_1, x_2)$ illustrated in Figure 6.2. The graph of the function is now a curved surface. Just as the curved line in two dimensional space corresponds to a curved surface in three dimensional space, the tangent line in two dimensional space corresponds to a tangent plane in three dimensional space. In chapter 2 on vectors I showed that any plane

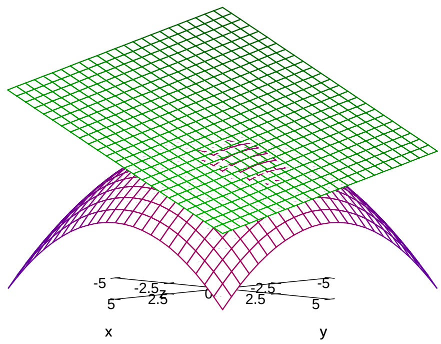

Figure 6.2: Appoximating the function at $(1, 2)$ $f(x) = 100 - x_1^2 - x_2^2$ at $(1, 2)$ by the plane $p(x) = 95 - 2(x_1 - 1) - 4(x_2 - 2)$.

through the point (x_{01}, x_{02}, y_0) can be written in vector notation as

$$(p_1, p_2, p_3)\left(\begin{pmatrix} x_1 \\ x_2 \\ y \end{pmatrix} - \begin{pmatrix} x_{01} \\ x_{02} \\ y_0 \end{pmatrix}\right) = 0$$

or equivalently

$$p_1(x_1 - x_{01}) + p_2(x_2 - x_{02}) + p_3(y - y_0) = 0$$

If $p_3 \neq 0$ this becomes

$$y = y_0 + a_1(x_1 - x_{01}) + a_2(x_2 - x_{02})$$

where $a_1 = -\frac{p_1}{p_3}$ and $a_2 = -\frac{p_2}{p_3}$. By analogy with the equation of the tangent line as $y = f(x_0) + f'(x_0)(x - x_0)$ the equation of the tangent plane should be

$$y = f(x_{01}, x_{02}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_1}(x_1 - x_{01}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_2}(x_2 - x_{02})$$

and the approximation should be

$$f(x_1, x_2) \approx f(x_{01}, x_{02}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_1}(x_1 - x_{01}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_2}(x_2 - x_{02}).$$

This suggests that the extension to n variables should be the approximation

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}) \text{ when } \mathbf{x} \text{ is close to } \mathbf{x}_0.$$

The argument I have just given is in no way rigorous, but it can be made so. This is done by assuming that the function $f(\mathbf{x})$ can be approximated by a function of the form

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} a_i (x_i - x_{i0})$$

and then showing that a_i must be the partial derivative $\frac{\partial f(\mathbf{x}_0)}{\partial x_i}$.

You have to be slightly careful with this, because unlike the situation with functions of a single variable it is possible that the derivatives exist but the approximation does not work. As an example consider the function $f : \mathcal{R}^2 \to \mathcal{R}$ given by

$$\begin{aligned} f(x_1, x_2) &= 0 \text{ when } x_1 x_2 = 0 \\ f(x_1, x_2) &= 1 \text{ when } x_1 x_2 \neq 0. \end{aligned}$$

This function is 0 when one or both of x_1 and x_2 are zero, and 1 otherwise. Thus $f(x_1, 0) = 0$ for all values of x_1 and $f(0, x_2) = 0$ for all values of x_2, implying that the partial derivatives

$$\frac{\partial f(0,0)}{\partial x_1} = \frac{\partial f(0,0)}{\partial x_2} = 0.$$

This suggests that when (x_1, x_2) is close to $(0, 0)$

$$f(0,0) + \frac{\partial f(0,0)}{\partial x_1} x_1 + \frac{\partial f(0,0)}{\partial x_2} x_2 \tag{6.1}$$

should be a good approximation to $f(x_1, x_2)$. However

$$f(0,0) = \frac{\partial f(0,0)}{\partial x_1} = \frac{\partial f(0,0)}{\partial x_2} = 0$$

so this implies that 0 is a good approximation to $f(x_1, x_2)$ when (x_1, x_2) is close to 0. But this is very far from being a good approximation because $f(x_1, x_2) = 1$ whenever $x_1 x_2 \neq 0$.

An assumption that ensures that the approximation works for a function $f : \mathcal{R}^{++} \to \mathcal{R}$ where

$$\mathcal{R}^{++} = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n,\ x_i > 0 \text{ for } i = 1, 2...n\}$$

is that f has partial derivatives on $\mathcal{R}^{++}$, and that the partial derivatives are continuous. To understand what this means you need to know what continuity means. Intuitively a function $g(\mathbf{x})$ is continuous at $\mathbf{x}_0$ if $g(\mathbf{x})$ gets closer to $g(\mathbf{x}_0)$ as $\mathbf{x}$ gets closer and closer to $\mathbf{x}_0$. More formally the function is continuous at $\mathbf{x}_0$ if $g(\mathbf{x}_0)$ is the limit of $g(\mathbf{x})$ as $\mathbf{x}$ tends to $\mathbf{x}_0$.

In order to avoid writing out the conditions many times I will use the following definition.

Definition 41 *The function $f(\mathbf{x}) : \mathcal{R}^{++} \to \mathcal{R}$ is a **well behaved** if it has partial derivatives, and the partial derivatives are continuous on $\mathcal{R}^{++}$.*

If a function is well behaved the approximation

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i} (x_i - x_{i0}) \text{ when } \mathbf{x} \text{ is close to } \mathbf{x}_0$$

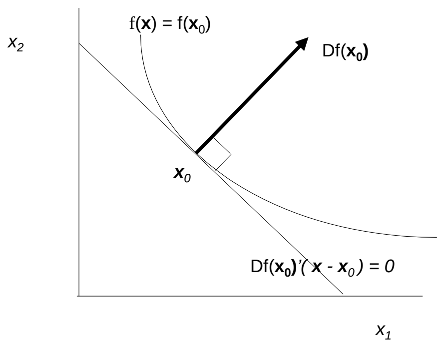

Figure 6.3: The Level Sets, Tangent and Partial Derivative Vector for a Function

works. Making this argument rigorous requires a precise definition of what is meant by approximation, and then a proof that existence and continuity of the partial derivatives implies that the approximation is valid.

The result is more general than that stated here. The set $\mathcal{R}^{++}$ can be replaced by an open subset of $\mathcal{R}^n$. See chapter, 8, but remember that this chapter is intended for MRes students and those interested in taking advanced micro (EC487). In practice economists almost always works with $\mathcal{R}^{++}$.

6.3 Level Sets and Vectors of Partial Derivatives

6.3.1 The Level Set and Tangent for One Function

A level set of a function $f(\mathbf{x})$is a set on which $f(\mathbf{x})$ is constant. Being formal about this:

Definition 42 *If S is a subset of $\mathcal{R}^n$ and the function $f: S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) = y\}$$

*of elements of S for which $f(\mathbf{x}) = y$ is called a **level set**.*

A level set of a utility function is an indifference curve; a level set of a production function is an isoquant. Now think about two points $\mathbf{x}$ and $\mathbf{x}_0$ that lie in the same level set so

$$f(\mathbf{x}) = f(\mathbf{x}_0)$$

and assume that the function $f: S \to \mathcal{R}$ is well behaved so close to $\mathbf{x_0}$

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}).$$

As $f(\mathbf{x}) = f(\mathbf{x}_0)$ this implies that

$$\sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}) \approx 0. \tag{6.2}$$

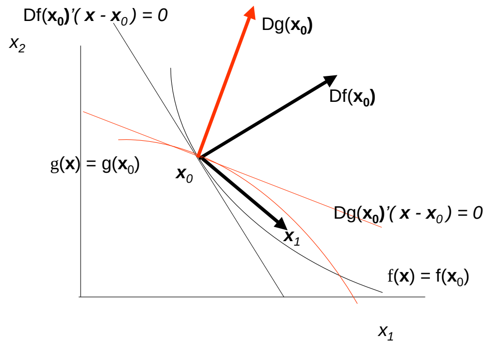

Figure 6.4: Level Sets and Partial Derivative Vectors for Two Functions

If I use notation

$$Df(\mathbf{x}_0) = \begin{pmatrix} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_2} \\ . \\ . \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \end{pmatrix}$$

so $Df(\mathbf{x}_0)$ is the n vector of partial derivatives equation 6.2 becomes

$$Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) \approx 0.$$

Thus when $\mathbf{x}$ is very close to $\mathbf{x}_0$ and $f(\mathbf{x}) = f(\mathbf{x}_0)$ $\mathbf{x}$ very nearly satisfies the equation.

$$Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) = 0.$$

This is an equation of the type that I discussed at some length in section 2.7 of chapter 2 on vectors. If $n = 2$ this is a straight line, if $n = 3$ it is a plane, and for $n > 3$ it is a hyperplane. In every case the vector of partial derivatives $Df(\mathbf{x}_0)$ is orthogonal (at 90°) to the line, plane or hyperplane. This is illustrated in Figure 8.2, which shows the vector of partial derivatives $Df(\mathbf{x}_0)$ which is orthogonal to the line $Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) = 0$. The line is a very good approximation to the level set when $\mathbf{x}$ is close to $\mathbf{x}_0$, and it seems reasonably intuitive and is in fact true that this line must be tangent to the level set.

6.3.2 Level Sets and Tangents for Two Functions

This section gives you a peek at the intuition behind Lagrangians which you will spend a lot of time studying in maths for micro and use extensively throughout your course. Figure 8.4 is the same as Figure 8.2 except that I have introduced another function $g(\mathbf{x})$, its partial derivative vector $Dg(\mathbf{x}_0)$ at $\mathbf{x}_0$ and its level set $g(\mathbf{x}) = g(\mathbf{x}_0)$. I have chosen to show the case where the two partial derivative vectors point in different directions. As the figure suggests this implies that the level sets cross each other so there is a point $\mathbf{x}_1$ which satisfies both $f(\mathbf{x}_1) > f(\mathbf{x}_0)$ and $g(\mathbf{x}_1) < g(\mathbf{x}_0)$. This tells you that $\mathbf{x}_0$ does not maximize $f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq g(\mathbf{x}_0)$. The only possible solutions are those in which the two partial derivative vectors $Df(\mathbf{x}_0)$ and $Dg(\mathbf{x}_0)$ point in the same direction; this requires that $Df(\mathbf{x}_0) = \lambda Dg(\mathbf{x}_0)$ where $\lambda \geq 0$. Written out component by component this requires that

$$\begin{aligned} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} &= \lambda \frac{\partial g(\mathbf{x}_0)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_2} &= \lambda \frac{\partial g(\mathbf{x}_0)}{\partial x_2} \end{aligned}$$

These are of course the first order conditions which come from differentiating the Lagrangian

$$\mathcal{L} = f(\mathbf{x}) + \lambda [c - g(\mathbf{x})].$$

The condition that $\lambda \geq 0$ comes from thinking about an inequality problem. You will come to think of it as the nonnegative multiplier condition of the Kuhn-Tucker theorem.

Chapter 7

Working with Multivariate Calculus

7.1 Partial Derivatives

In single variable calculus the derivative $f'(x)$ of a function $f(x)$ is defined as the limit as h tends to 0 of

$$\frac{f(x+h)-f(x)}{h}.$$

The corresponding idea for multivariate calculus is the partial derivative; if $f(\mathbf{x})$ is a function of an n vector $\mathbf{x}=(x_1,x_2,...x_n)$ the partial derivative of f with respect to x_i is the limit as h_i tends to 0 of

$$\frac{f(x_1,x_2,...x_i+h_i...x_n)-f(x_1,x_2,...x_i...x_n)}{h_i}.$$

The notation for the partial derivative is either $f_i(x_1,x_2,...x_i...x_n)$ or

$$\frac{\partial f(x_1,x_2,...x_i...x_n)}{\partial x_i}$$

these expressions can be written more concisely as $f_i(\mathbf{x})$ or $\dfrac{\partial f(\mathbf{x})}{\partial x_i}$. There are issues about the existence of partial derivatives, just as there are issues about the existence of derivatives; for example the function $\min(x_1,x_2)$ does not have partial derivatives at $x_1=x_2$.

With one very important exception, the chain rule, you can use the rules for differentiation (sum, product, and quotient rules) when finding partial derivatives in exactly the same way as you use the rules finding the derivative of a function of a single variable. When finding $\frac{\partial f(\mathbf{x})}{\partial x_i}$ you simply treat $x_1,x_2,...x_{i-1},x_{i+1}...x_n$ as constants. However the chain rule is more complicated because it has to cope with the situation in which several of the arguments of a function depend upon another set of variables. I now turn to the chain rule.

7.2 The Chain Rule for Partial Derivatives

7.2.1 Where the Rule Comes From

This is essential material because you will need to use the chain rule extensively, particularly in your microeconomics course. The chain rule for functions of a single variable says that if $g(x) = f(z(x))$ then

$$\frac{dg(x)}{dx} = \frac{df(z(x))}{dz}\frac{dz(x)}{dx}.$$

The reason the chain rule holds is that the essential point about a differentiable function $f(z)$ of a single variable z is that

$$f(z) \approx f(z_0) + \frac{df(z_0)}{dz}(z - z_0)$$

when z is close to z_0. Similarly

$$z(x) \approx z(x_0) + \frac{dz(x_0)}{dx}(x - x_0)$$

when x is close to x_0. Thus if $z(x_0) = z_0$

$$g(x) - g(x_0) = f(z(x)) - f(z(x_0)) \approx \frac{df(z_0)}{dz}(z(x) - z(x_0)) \approx \frac{df(z_0)}{dz}\frac{dz(x_0)}{dx}(x - x_0)$$

so

$$\frac{g(x) - g(x_0)}{(x - x_0)} \approx \frac{df(z_0)}{dz}\frac{dz(x_0)}{dx}.$$

Taking limits gives the result. The argument as I have given it cannot be made fully rigorous because of the possibility that $z(x) - z(x_0) = 0$ at some point. Putting this right requires the assumption that the derivatives are continuous.

The chain rule for functions of many variables says that if $g(\mathbf{x}) = f(\mathbf{z}(\mathbf{x}))$ where $x = (x_1, x_2, ...x_n)$ and $z(\mathbf{x})$ is the vector of functions $(z_1(\mathbf{x}), z_2(\mathbf{x}) ... z_m(\mathbf{x}))$

$$\frac{\partial g(\mathbf{x})}{\partial x_i} = \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}))}{\partial z_j}\frac{\partial z_j(\mathbf{x})}{\partial x_i}.$$

The rule requires that the functions $f(\mathbf{z})$ and $z_j(\mathbf{x})$ $(j = 1, 2...m)$ have continuous partial derivatives.

The reason the chain rule holds is that the essential point about a differentiable function $f(\mathbf{z})$ of a vector of variables $z = (z_1, z_2 ... z_m)$ is that

$$f(\mathbf{z}) \approx f(\mathbf{z}_0) + \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}_0)}{\partial z_j}(z_j - z_{0j})$$

when z is close to z_0. Similarly if x_i varies, and all the other components of x are constant

$$z_j(\mathbf{x}) \approx z_j(\mathbf{x}_0) + \frac{\partial z_j(\mathbf{x}_0)}{\partial x_i}(x_i - x_{0i})$$

when x is close to x_0. Thus if $z_0 = z(\mathbf{x}_0)$

$$\begin{aligned}
g(\mathbf{x}) - g(\mathbf{x}_0) &= f(\mathbf{z}(\mathbf{x})) - f(\mathbf{z}_0) \\
&\approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j} (z_j(\mathbf{x}) - z_j(\mathbf{x}_0)) \\
&\approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j} \frac{\partial z_j(\mathbf{x}_0)}{\partial x_i} (x_i - x_{0i})
\end{aligned}$$

so

$$\frac{g(\mathbf{x}) - g(\mathbf{x}_0)}{(x_i - x_{0i})} \approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j} \frac{\partial z_j(\mathbf{x}_0)}{\partial x_i}$$

Taking limits gives the result.

This argument is suggestive rather than fully rigorous both because of the possibility that $u_j(v) - u_j(v_0) = 0$ at some point, and because as I argued in the previous chapter, the approximation of the graph of the function by a plane requires continuity of partial derivatives. In practice economists almost always assume continuity of partial derivatives, usually without saying so.

7.2.2 Using the Chain Rule: an Example

Suppose that $f(z_1, z_2) = z_1^{1/4} z_2^{3/4}$. Let $g(x_1, x_2) = z_1^{1/4} z_2^{3/4}$ where $z_1(x_1, x_2) = 2x_1 + 3x_2$ and $z_2(x_1, x_2) = 4x_1 + 5x_2$. Then using the chain rule, differentiating and then substituting

$$\begin{aligned}
\frac{\partial g(x_1, x_2)}{\partial x_1} &= \frac{\partial f(z_1, z_2)}{\partial z_1} \frac{\partial z_1(x_1, x_2)}{\partial x_1} + \frac{\partial f(z_1, z_2)}{\partial z_2} \frac{\partial z_2(x_1, x_2)}{\partial x_1} \\
&= \frac{\partial f(z_1, z_2)}{\partial z_1} 2 + \frac{\partial f(z_1, z_2)}{\partial z_2} 4 \\
&= \frac{1}{4} z_1^{-3/4} z_2^{3/4} 2 + \frac{3}{4} z_1^{1/4} z_2^{-1/4} 4 \\
&= \frac{1}{2} \left(\frac{z_2}{z_1}\right)^{3/4} + 3 \left(\frac{z_1}{z_2}\right)^{1/4} \\
&= \frac{1}{2} \left(\frac{(4x_1 + 5x_2)}{(2x_1 + 3x_2)}\right)^{3/4} + 3 \left(\frac{(2x_1 + 3x_2)}{(4x_1 + 5x_2)}\right)^{1/4}.
\end{aligned}$$

As a check on the differentiation you can also substitute to get

$$g(x_1, x_2) = (2x_1 + 3x_2)^{1/4} (4x_1 + 5x_2)^{3/4}$$

and then differentiate using the product rule to get

$$\begin{aligned}
\frac{\partial g(x_1, x_2)}{\partial x_1} &= \frac{2}{4} (2x_1 + 3x_2)^{-3/4} (4x_1 + 5x_2)^{3/4} + 3 (2x_1 + 3x_2)^{1/4} (4x_1 + 5x_2)^{-1/4} \\
&= \frac{1}{2} \left(\frac{(4x_1 + 5x_2)}{(2x_1 + 3x_2)}\right)^{3/4} + 3 \left(\frac{(2x_1 + 3x_2)}{(4x_1 + 5x_2)}\right)^{1/4}.
\end{aligned}$$

7.2.3 The Chain Rule with Overlapping Variables: an Example

This is a very important example, because many problems in consumer theory are similar. The previous example worked with functions of two variables z_1 and z_2 which are themselves functions of two other different variables x_1 and x_2. In fact the two sets of variables often overlap as in this example.

You will learn that a consumer maximizing a Cobb-Douglas utility function $x_1^{1/3}x_2^{2/3}$ subject to a budget constraint $p_1x_1 + p_2x_2 \leq m$, where p_1 and p_2 are prices, and m is the amount of money the consumer has, buys amounts of the goods

$$\begin{aligned} x_1(p_1, p_2, m) &= \frac{1}{3}\frac{m}{p_1} \\ x_2(p_1, p_2, m) &= \frac{2}{3}\frac{m}{p_2}. \end{aligned}$$

Now suppose that $m = p_1\omega_1 + p_2\omega_2$. Economically this is the case where the consumer has a given endowment (ω_1, ω_2) rather than a given amount of money m. This is the situation in general equilibrium theory. Let

$$\begin{aligned} k_1(p_1, p_2, \omega_1, \omega_2) &= x_1(p_1, p_2, m) \\ k_2(p_1, p_2, \omega_1, \omega_2) &= x_2(p_1, p_2, m) \end{aligned}$$

when $m = p_1\omega_1 + p_2\omega_2$. You can think of the variables of the original function p_1, p_2 and m being functions of the new variables p_1, p_2, ω_1 and ω_2. The functions are

$$\begin{aligned} p_1 &= p_1 \\ p_2 &= p_2 \\ m &= p_1\omega_1 + p_2\omega_2. \end{aligned}$$

Using that chain rule

$$\begin{aligned} \frac{\partial k_1(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1}\frac{\partial p_1}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial p_2}\frac{\partial p_2}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\frac{\partial m}{\partial p_1} \\ &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\omega_1 \end{aligned}$$

because $\dfrac{\partial p_1}{\partial p_1} = 1$, $\dfrac{\partial p_2}{\partial p_1} = 0$ and $\dfrac{\partial m}{\partial p_1} = \omega_1$. Once you are familiar with using the chain rule in this way you can omit the first line and write down at once

$$\begin{aligned} \frac{\partial k_1(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\frac{\partial m}{\partial p_1} \\ &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\omega_1. \end{aligned}$$

As $x_1(p_1, p_2, m) = \dfrac{1}{3}\dfrac{m}{p_1}$, $\dfrac{\partial x_1(p_1, p_2, m)}{\partial p_1} = -\dfrac{1}{3}\dfrac{m}{p_1^2}$ and $\dfrac{\partial x_1(p_1, p_2, m)}{\partial m} = \dfrac{1}{3p_1}$

so

$$
\begin{aligned}
\frac{\partial k_1(p_1,p_2,\omega_1,\omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1,p_2,m)}{\partial p_1} + \frac{\partial x_1(p_1,p_2,m)}{\partial m}\frac{\partial m}{\partial p_1} \\
&= -\frac{1}{3}\frac{m}{p_1^2} + \frac{\omega_1}{3p_1} \\
&= -\frac{1}{3}\frac{p_1\omega_1 + p_2\omega_2}{p_1^2} + \frac{\omega_1}{3p_1} \\
&= \frac{1}{3p_1^2}(p_1\omega_1 - p_1\omega_1 - p_2\omega_2) \\
&= -\frac{p_2\omega_2}{3p_1^2}.
\end{aligned}
$$

As a check $m = p_1\omega_1 + p_2\omega_2$ so

$$k_1(p_1,p_2,\omega_1,\omega_2) = \frac{p_1\omega_1 + p_2\omega_2}{3p_1} = \frac{\omega_1}{3} + \frac{p_2\omega_2}{3p_1}$$

implying that

$$\frac{\partial k_1(p_1,p_2,\omega_1,\omega_2)}{\partial p_1} = -\frac{p_2\omega_2}{3p_1^2}.$$

In this case it is much easier to do the substitution before differentiating, rather than using the chain rule. The example is here to help you become familiar with the chain rule for partial derivatives, which is important in other contexts.

7.2.4 Notation for the Chain Rule

Notation gets particularly difficult when variables overlap. The simplest case is when the function $g(x_1)$ is defined as

$$g(x_1) = f(x_1, x_2(x_1))$$

where x_2 is a function of x_1. Using the chain rule to differentiate $g(x_1)$ gives

$$\frac{dg(x_1)}{dx_1} = \frac{\partial f(x_1,x_2)}{\partial x_1} + \frac{\partial f(x_1,x_2)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}.$$

Because $g(x_1)$ and $x_2(x_1)$ are functions of a single variable x_1 the derivatives $\frac{dg(x_1)}{dx_1}$ and $\frac{dx_2(x_1)}{dx_1}$ are ordinary and not partial derivatives. Sometimes $\frac{\partial f(x_1,x_2)}{\partial x_1}$ is called the partial derivative of f with respect to x_1 and $\frac{\partial f(x_1,x_2)}{\partial x_1} + \frac{\partial f(x_1,x_2)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}$ is called the total derivative of f with respect to x_1.

However things get more complicated when there are more variables, for example if

$$g(x_1,x_3) = f(x_1, x_2(x_1), x_3).$$

Then

$$\frac{\partial g(x_1,x_3)}{\partial x_1} = \frac{\partial f(x_1,x_2,x_3)}{\partial x_1} + \frac{\partial f(x_1,x_2,x_3)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}.$$

The term $\dfrac{\partial g(x_1, x_3)}{\partial x_1}$ is now a partial derivative because x_3 is being held constant. Using the term total derivative in this context may be confusing.

Different people have different ways of dealing with these notational difficulties. Once you are thoroughly familiar with using the chain rule the notation stops causing confusion, but whilst you are learning you may find it helpful to work in the following way.

I defined the functions $k_1(p_1, p_2, \omega_1, \omega_2)$ in the previous section by writing that $k_1(p_1, p_2, \omega_1, \omega_2) = x_1(p_1, p_2, m)$ when $m = p_1\omega_1 + p_2\omega_2$. It is tempting to consider simply the function $x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)$. But it causes problems if you want to differentiate. What do you mean by $\dfrac{\partial x_1}{\partial p_1}$? You could find yourself writing

$$\frac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial p_1} = \frac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial m}\omega_1$$

where on the left hand side $\dfrac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial p_1}$ is in fact $\dfrac{\partial k_1(p_1, p_2, \omega_1, \omega_2)}{\partial p_1}$ and on the right hand side $\dfrac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial p_1}$ is in fact $\dfrac{\partial x_1(p_1, p_2, m)}{\partial p_1}$ evaluated at $m = p_1\omega_1 + p_2\omega_2$. You may also be puzzled by what

$$\frac{\partial x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)}{\partial m}$$

means, since m does not appear in $x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)$.

You may find it helpful to follow these rules:

- Use different notation for different functions. Here $k_1(p_1, p_2, \omega_1, \omega_2)$ and $x_1(p_1, p_2, m)$ both give the quantity of good 1, but they give the quantity as a function of different things.
- Write out all the arguments of functions, writing for example $x_1(p_1, p_2, m)$ rather than x_1. You can use vector notation, for example writing $\mathbf{p}$ for p_1, p_2.
- When you define a new function do it by writing that $k_1(p_1, p_2, \omega_1, \omega_2) = x_1(p_1, p_2, m)$ when $m = p_1\omega_1 + p_2\omega_2$ rather than writing $k_1(p_1, p_2, \omega_1, \omega_2) = x_1(p_1, p_2, p_1\omega_1 + p_2\omega_2)$.

Of course following these rules makes writing things out a lot slower, and puts more on the page, which makes it visually harder to follow what is happening. Once you are thoroughly familiar with handling partial derivatives you can follow the rules in your head without writing everything down. But if an argument made with partial derivatives puzzles you it can be very helpful to write out the argument using these rules.

7.3 Directional Derivatives

Directional derivatives are an application of the chain rule. Consider the differentiable function $f : \mathcal{R}^n \to \mathcal{R}$, think of $\mathbf{x}_0$ and $\mathbf{x}_1$ as fixed n vectors and t as a number that varies and let

$$x_i = x_{0i} + t\left(x_{1i} - x_{0i}\right).$$

or in vector notation

$$\mathbf{x} = \mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right).$$

Note that this implies that

$$\frac{\partial x_i}{\partial t} = \left(x_{1i} - x_{0i}\right). \tag{7.1}$$

Let

$$g\left(t, \mathbf{x}_0, \mathbf{x}_1\right) = f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right). \tag{7.2}$$

The using the chain rule and equation 7.1

$$\begin{aligned} \frac{\partial g\left(t, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} &= \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right)}{\partial x_i} \frac{\partial x_i}{\partial t} \\ &= \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right)}{\partial x_i} \left(x_{1i} - x_{0i}\right). \end{aligned}$$

In particular at $t = 0$ this becomes

$$\frac{\partial g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} = \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_i} \left(x_{1i} - x_{0i}\right). \tag{7.3}$$

This is the directional derivative, it represents the change in the function when you move from $\mathbf{x}_0$ along the line $\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x_0}\right)$ holding $\mathbf{x}_0$ and $\mathbf{x}_1$ constant and varying.

7.4 Local Maxima and Minima

The most basic result on maximization and partial derivatives is.

Proposition 43 *Suppose that $f : \mathcal{R}^{++} \to \mathcal{R}$ is differentiable, with continuous partial derivatives, and $\mathbf{x_0}$ is a local maximum or minimum of $f(\mathbf{x})$, that is $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ when $\mathbf{x}$ is close to $\mathbf{x}_0$. Then*

$$\frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_i} = 0 \textit{ for } i = 1, 2...n.$$

The reason behind this is the same as the reason behind the corresponding result for maximization of a function of a single variable. For such a function $f(x)$, if the derivative $df(x_0)/dx > 0$ the function is increased by increasing x slightly and decreased by decreasing x slightly so the function does not have a maximum or minimum at x_0. If the derivative $df(x_0)/dx < 0$ the function is decreased by increasing x slightly and increased by decreasing x slightly so the function does not have a maximum or minimum at x_0. Similarly if $\partial f(\mathbf{x}_0)/\partial x_i \neq 0$ the function can be increased and decreased by a small change in x_i.

7.5 Homogeneous Functions

7.5.1 Definition

A function $f(x_1, x_2, ...x_n)$ is homogeneous of degree k if for any $t > 0$

$$f(tx_1, tx_2, ...tx_n) = t^k f(x_1, x_2, ...x_n). \tag{7.4}$$

Some functions are homogeneous in some but not all variables. For example, you will meet the expenditure function $e(p_1, p_2...p_n, u)$ in consumer theory. The expenditure function $e(p_1, p_2...p_n, u)$ is a function of prices $p_1, p_2...p_n$ and utility u. The expenditure function is homogeneous of degree 1 in prices, that is

$$e(tp_1, tp_2...tp_n, u) = te(p_1, p_2...p_n, u).$$

7.5.2 Homogeneous function and derivatives

The chain rule is very useful in proving results about homogeneous functions.

Proposition 44 *If $f(x_1, x_2, ...x_n)$ is homogeneous of degree k that is*

$$f(tx_1, tx_2, ...tx_n) = t^k f(x_1, x_2, ...x_n)$$

then $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$ is homogeneous of degree $k-1$, that is, using notation $f_i(x_1, x_2, ...x_n)$ for $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$

$$f_i(tx_1, tx_2, ...tx_n) = t^{k-1} f_i(x_1, x_2, ...x_n).$$

Proof. Let $z_i = tx_i$ for all i so equation 7.4 becomes

$$f(z_1, z_2, ...z_n) = t^k f(x_1, x_2, ...x_n)$$

where $z_i = tx_i$ for $i = 1, 2...n$. As this equation holds for all $(x_1, x_2, ...x_n)$ the derivatives of the two sides are equal. The derivative of the left hand side with respect to x_i is

$$\frac{\partial f(z_1, z_2, ...z_n)}{\partial z_i} \frac{\partial z_i}{\partial x_i} = \frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i} t.$$

The derivative of the right hand side with respect to x_i is

$$t^k \frac{f(x_1, x_2, ...x_n)}{\partial x_i}$$

Equating the two sides

$$\frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i} t = t^k \frac{f(x_1, x_2, ...x_n)}{\partial x_i}$$

or using notation $f_i(x_1, x_2, ...x_n)$ for $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$

$$f_i(tx_1, tx_2, ...tx_n) = t^{k-1} f_i(x_1, x_2, ...x_n).$$

(The notation gets awkward here. It is tempting to write the left hand side as $\dfrac{\partial f(tx_1, tx_2, ...tx_n)}{\partial x_i}$ but that is not the same as $\dfrac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i}$ which is what is wanted). ■

This is a useful result. You may for example be interested in the marginal rate of technical substitution between two inputs i and j at two points $(x_1, x_2, ...x_n)$ and $(tx_1, tx_2, ...tx_n)$. These are given by

$$\frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)} \quad \text{and} \quad \frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)}$$

If f is homogeneous of degree k so $f_i(x_1, x_2, ...x_n)$ is homogeneous of degree $k-1$ so

$$\frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)} = \frac{t^{k-1} f_j(x_1, x_2, ...x_n)}{t^{k-1} f_i(x_1, x_2, ...x_n)} = \frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)}$$

and

$$\frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)} = \frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)}.$$

This says that the marginal rate technical of substitution is constant along a ray from the origin in the isoquant diagram.

7.6 Constant returns to scale and Euler's Theorem

You will use the result which is called Euler's Theorem in macroeconomics. (Euler was a very important eighteenth century mathematician, so a number of results carry his name.)

Definition 45 *A production function $y = f(x_1, x_2, ...x_n)$ which gives output y as a function of inputs $(x_1, x_2, ...x_n)$ displays constant returns to scale if it is homogeneous of degree 1, that is for any positive number t*

$$f(tx_1, tx_2, ...tx_n) = tf(x_1, x_2, ...x_n).$$

The way to think about constant returns to scale is that changing all inputs by the same proportion changes output by the same proportion.

Theorem 46 *Euler's Theorem states that if $f : \mathcal{R}^{++} \to \mathcal{R}^{+}$ and $f(x_1, x_2, ...x_n)$ is homogeneous of degree one then*

$$\sum_{i=1}^{n} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i = f(x_1, x_2, ...x_n).$$

Proof. *The proof is another example of using the chain rule to do proofs. This is an important and useful technique. The key is that you must have a relationship that holds for all values of the variables in a relevant set. In this* *and many other cases in economics this is the set of vectors whose components are all non-negative. In this case the relationship is*

$$f\left(tx_1, tx_2, ...tx_n\right) = tf\left(x_1, x_2, ...x_n\right). \tag{7.5}$$

Because this relationship holds for all values of the variables, if you find the partial derivative of both sides of the equation with respect to the same variable, the partial derivatives must be equal. Differentiating both sides by t gives

$$\frac{\partial f\left(tx_1, tx_2, ...tx_n\right)}{\partial t} = f\left(x_1, x_2, ...x_n\right). \tag{7.6}$$

To find $\dfrac{\partial f\left(tx_1, tx_2, ...tx_n\right)}{\partial t}$ *let $z_i = tx_i$ so using the chain rule*

$$\begin{aligned}\frac{\partial f\left(z_1, z_2, ...z_n\right)}{\partial t} &= \sum_{i=1}^{n} \frac{\partial f\left(z_1, z_2, ...z_n\right)}{\partial z_i}\frac{\partial z_i}{\partial t} \\ &= \sum_{i=1}^{n} \frac{\partial f\left(z_1, z_2, ...z_n\right)}{\partial z_i} x_i.\end{aligned}$$

Letting $t = 1$ so $z_i = x_i$ this becomes

$$\frac{\partial f\left(x_1, x_2, ...x_n\right)}{\partial t} = \sum_{i=1}^{n} \frac{\partial f\left(x_1, x_2, ...x_n\right)}{\partial x_i} x_i.$$

Substituting this expression in the identity 7.6 the identity becomes

$$\sum_{i=1}^{n} \frac{\partial f\left(x_1, x_2, ...x_n\right)}{\partial x_i} x_i = f\left(x_1, x_2, ...x_n\right).$$

■

The result is economically important, because if a firm is a price taker in both its output and input markets and the price of the output is p and the price of input i is w_i the first order conditions for profit maximization imply that

$$p\frac{\partial f\left(x_1, x_2, ...x_n\right)}{\partial x_i} = w_i$$

so taken with Euler's theorem this implies that

$$pf\left(x_1, x_2, ...x_n\right) = \sum_{i=1}^{n} w_i x_i$$

so the firm makes zero profits. The revenue from production is all distributed to the factors of production. This is a condition relating prices to the technology of the firm. The condition is required for an equilibrium in which the firm is a price taker. If it is not satisfied there are two possibilities. One is that the firm makes losses at any positive level of output so it shuts down. The other is that it can make strictly positive profits at some level of output, so with constant returns to scale it can expand output to make infinitely large profits so long as it is a price taker. At some stage it will have a large market share and it will no longer be a price taker.

Production functions are also used in growth theory in macroeconomics, where the function gives the output of an entire economy as a function of inputs, often capital and labour. In this context Euler's theorem says that the entire national income is distributed to the factors of production.

7.7 Second derivatives and Young's Theorem

If $f(\mathbf{x})$ is a function of $\mathbf{x} = (x_1, x_2, ...x_n)$ the second partial derivative of the $f(\mathbf{x})$ with respect to x_i is defined analogously to the second derivative of a function of a single variable so

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i^2} = \frac{\partial \left(\dfrac{\partial f(\mathbf{x})}{\partial x_i} \right)}{\partial x_i}.$$

Functions of many variables also have cross partial derivatives

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial \left(\dfrac{\partial f(\mathbf{x})}{\partial x_i} \right)}{\partial x_j}.$$

It is not always true that

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}.$$

But it is true that

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$$

if $f(\mathbf{x})$ has continuous second derivatives. This is Young's theorem. The matrix of second derivatives

$$\left[\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} \right]$$

is called the Hessian matrix.

7.8 The Implicit Function Theorem

Implicit functions are important and familiar in economics. You will be familiar with drawing indifference curves in consumer theory. These show x_2 as a function of x_1, but are derived from the equation $u(x_1, x_2) = u_1$ for some fixed number u_1. Different indifference curves correspond to different values u_1, u_2.... Sometimes you can derive x_2 as an explicit function of x_1 and u_1 by rearranging the equation $u(x_1, x_2) = u_1$, sometimes this is difficult, or impossible. But this does not fundamentally matter if you can be sure the function exists and can find its derivatives.

Existence can be a problem. You want to find the function $y = f(x)$ defined implicitly by the equation

$$x^2 + y^2 = 25.$$

If you solve this equation for y you get solutions

$$y = \sqrt{(25 - x^2)}$$

and

$$y = -\sqrt{(25 - x^2)}.$$

There is a unique solution $y = 0$ if $x = 5$ or $x = -5$. If $x < -5$ or $5 < x$ there are no real valued solutions and if $-5 < x < 5$ there are two solutions. By definition a function of x has a unique value for each value of x. The most that can be asked for is uniqueness in a neighbourhood. For example if (x, y) has to be is close to $(3, 4)$ the only is solution $y = \sqrt{(25 - x^2)}$. Even so there can be problems. There are two solutions in any neighbourhood of $(5, 0)$. The implicit function theorem gives conditions under which an implicit function exists, and shows how to find its derivative.

Theorem 47 (The implicit function theorem) *Suppose that* $G(x_1, x_2, ..x_k, y)$ *is a continuous function with continuous partial derivatives defined in a neighbourhood of a point* $(x_1^*, x_2^*, ..x_k^*, y^*)$, *with*

$$G(x_1^*, x_2^*, ..x_k^*, y^*) = c$$

and

$$\frac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial y} \neq 0.$$

Then there is function $y(x_1, x_2....x_k)$ *with continuous partial derivatives defined on a neighbourhood of* $(x_1^*, x_2^*, ..x_k^*)$ *with the property that*

$$G(x_1, x_2....x_k, y(x_1, x_2....x_k)) = c$$

$$y^* = y(x_1^*, x_2^*, ..x_k^*)$$

and

$$\frac{\partial y(x_1^*, x_2^*, ..x_k^*)}{\partial x_i} = -\frac{\dfrac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial x_i}}{\dfrac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial y}}.$$

The difficult part of the theorem is proving existence. The result on the derivative follows easily from the chain rule. Suppose that

$$G(x_1, x_2, ..x_k, y(x_1, x_2, ..x_k)) = c \tag{7.7}$$

where c is a constant, Using the chain rule the derivative of $G(x_1, x_2, ..x_k, y(x_1, x_2, ..x_k))$ with respect to x_i is

$$\frac{\partial G(x_1, x_2, ..x_k, y)}{\partial x_i} + \frac{\partial G(x_1, x_2, ..x_k, y)}{\partial y} \frac{\partial y(x_1, x_2, ..x_k)}{\partial x_i} = 0$$

as $\dfrac{\partial c}{\partial x_i} = 0$. Thus if $\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial y} \neq 0$

$$\frac{\partial y(x_1, x_2, ..x_k)}{\partial x_i} = -\frac{\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial x_i}}{\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial y}}.$$

7.9 Taylor's Expansion with Multivariate Calculus

7.9.1 Taylor's approximation with a single variable

In the chapter on Taylor series in the Background Notes I showed that a function $f(x)$ evaluated at x_0 can be approximated by a polynomial of degree m. At x_0 the value of the function and its first, second ....m^{th} derivatives are the same as the corresponding values of the polynomial. The polynomial is

$$f(x_0)+\sum_{k=1}^{m}\frac{f^k(x_0)}{k!}(x-x_0)^k.$$

This section shows how the result can be extended to functions of many variables, and in particular demonstrates that the quadratic approximation is

$$f(\mathbf{x})\approx f(\mathbf{x}_0)+\sum_{i=1}^{n}\frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_{1i}-x_{0i})+\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i\partial x_j}(x_i-x_{0i})(x_j-x_{0j}).$$

The right hand side of this relation is a quadratic form that can be written in vector notation as

$$f(\mathbf{x})\approx f(\mathbf{x}_0)+\left(\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{x}}\right)'(\mathbf{x}-\mathbf{x}_0)+\frac{1}{2}(\mathbf{x}-\mathbf{x}_0)'\frac{\partial^2 f(\mathbf{x}_0)}{\partial \mathbf{x}^2}(\mathbf{x}-\mathbf{x}_0)$$

where $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{x}}$ is an n vector whose i^{th} component is $\frac{\partial f(\mathbf{x}_0)}{\partial x_i}$ and $\frac{\partial^2 f(\mathbf{x}_0)}{\partial \mathbf{x}^2}$ is an $n\times n$ matrix for which component ij is $\frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i\partial x_j}$.

7.9.2 Proving the Result

The result is proved using the same trick as I used when defining a directional derivative partial derivatives. This works with the function $g(t,\mathbf{x_0},\mathbf{x_1})$ defined as

$$g(t,\mathbf{x_0},\mathbf{x_1})=f(\mathbf{x}_0+t(\mathbf{x_1}-\mathbf{x_0}))$$

Thinking of $\mathbf{x}_0$ and $\mathbf{x}_0$ as fixed, and treating $g(t,\mathbf{x_0},\mathbf{x_1})$ as a function of a single variable t the polynomial approximation is

$$g(t,\mathbf{x_0},\mathbf{x_1})\approx g(0,\mathbf{x_0},\mathbf{x_1})+\sum_{k=1}^{m}\frac{1}{k!}\frac{\partial^k g(0,\mathbf{x_0},\mathbf{x_1})}{\partial t^k}t^k.$$

It is possible in principle to use the chain rule to write down the derivative

$$\frac{\partial^k g(0,\mathbf{x_0},\mathbf{x_1})}{\partial t^k}$$

in terms of f, $\mathbf{x}_0$, $\mathbf{x}_1$ and t. In practice economists confine themselves to the quadratic case with $m=2$,

$$g(t,\mathbf{x_0},\mathbf{x_1})\approx g(0,\mathbf{x_0},\mathbf{x_1})+t\frac{\partial g(0,\mathbf{x_0},\mathbf{x_1})}{\partial t}+\frac{1}{2}t^2\frac{\partial^2 g(0,\mathbf{x_0},\mathbf{x_1})}{\partial t^2}. \tag{7.8}$$

The next task is getting an expressions for the right hand side of this relation in terms of f, $\mathbf{x}_0$, $\mathbf{x}_1$ and t. From the definition of g

$$g\left(0,\mathbf{x_0},\mathbf{x_1}\right)=f\left(\mathbf{x}_0\right). \tag{7.9}$$

The chain rule implies that

$$\frac{\partial g\left(t,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t}=\sum_{j=1}^{n}\frac{\partial f\left(\mathbf{x}_0+t\left(\mathbf{x_1}-\mathbf{x_0}\right)\right)}{\partial x_j}\left(x_{1j}-x_{0j}\right) \tag{7.10}$$

and in particular that

$$\frac{\partial g\left(0,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t}=\sum_{j=1}^{n}\frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_{1j}-x_{0j}\right). \tag{7.11}$$

I now want the second derivative $\dfrac{\partial^2 g\left(t,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t^2}$, so I need to differentiate the right hand side of equation 7.10 with respect to t. Using the chain rule for partial derivatives the derivative of

$$\frac{\partial f\left(\mathbf{x}_0+t\left(\mathbf{x_1}-\mathbf{x_0}\right)\right)}{\partial x_j}$$

with respect to t is

$$\sum_{i=1}^{n}\frac{\partial^2 f\left(\mathbf{x}_0+t\left(\mathbf{x_1}-\mathbf{x_0}\right)\right)}{\partial x_i\partial x_j}\left(x_{1i}-x_{0i}\right)$$

so from equation 7.11

$$\frac{\partial^2 g\left(t,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t^2}=\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2 f\left(\mathbf{x}_0+t\left(\mathbf{x_1}-\mathbf{x_0}\right)\right)}{\partial x_i\partial x_j}\left(x_{1i}-x_{0i}\right)\left(x_{1j}-x_{0j}\right)$$

and in particular at $t=0$

$$\frac{\partial^2 g\left(0,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t^2}=\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial f^2\left(\mathbf{x}_0\right)}{\partial x_i\partial x_j}\left(x_{1i}-x_{0i}\right)\left(x_{1j}-x_{0j}\right) \tag{7.12}$$

Thus from equations 7.8, 7.9, 7.11 and 7.12

$$\begin{aligned}
& f\left(\mathbf{x}_0+t\left(\mathbf{x_1}-\mathbf{x_0}\right)\right) \\
= \; & g\left(t,\mathbf{x}_0,\mathbf{x}_1\right) && (7.13)\\
\approx \; & g\left(0,\mathbf{x}_0,\mathbf{x}_1\right)+t\frac{\partial g\left(0,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t}+\frac{1}{2}t^2\frac{\partial^2 g\left(0,\mathbf{x}_0,\mathbf{x}_1\right)}{\partial t^2} \\
= \; & f\left(\mathbf{x}_0\right)+t\left(\sum_{j=1}^{n}\frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_{1j}-x_{0j}\right)\right) \\
& +\frac{1}{2}t^2\left(\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2 f\left(\mathbf{x}_0\right)}{\partial x_i\partial x_j}\left(x_{1i}-x_{0i}\right)\left(x_{1j}-x_{0j}\right)\right). && (7.14)
\end{aligned}$$

Now let

$$\mathbf{x} = \mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)$$

so

$$\begin{aligned} g\left(t, \mathbf{x}_0, \mathbf{x}_1\right) &= f\left(\mathbf{x}\right) \\ t\left(x_{1j} - x_{0j}\right) &= x_j - x_{0j} \\ t^2\left(x_{1i} - x_{0i}\right)\left(x_{1j} - x_{0j}\right) &= \left(x_i - x_{0i}\right)\left(x_j - x_{0j}\right) \end{aligned}$$

and substituting in equation 7.14 gives the result

$$f\left(\mathbf{x}\right) \approx f\left(\mathbf{x}_0\right) + \sum_{j=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_j - x_{0j}\right) + \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial f^2\left(\mathbf{x}_0\right)}{\partial x_i \partial x_j}\left(x_i - x_{0i}\right)\left(x_j - x_{0j}\right). \tag{7.15}$$

This can also be written in vector and matrix notation. Let $Df\left(\mathbf{x}_0\right)$ be a column vector whose i^{th} component is $\frac{\partial f(\mathbf{x}_0)}{\partial x_j}$. Let $\mathbf{x} - \mathbf{x}_0$ be a column vector whose i^{th} component is $x_i - x_{i0}$ and let $D^2 f\left(\mathbf{x}_0\right)$ be an $n \times n$ matrix for which component ij is $\frac{\partial f^2(\mathbf{x}_0)}{\partial x_i \partial x_j}$. Then equation 7.15 becomes

$$f\left(\mathbf{x}_0\right) \approx f\left(\mathbf{x}_0\right) + Df\left(\mathbf{x}_0\right)'\left(\mathbf{x} - \mathbf{x_0}\right) + \frac{1}{2}\left(\mathbf{x} - \mathbf{x_0}\right)' D^2 f\left(\mathbf{x}_0\right)\left(\mathbf{x} - \mathbf{x_0}\right).$$

Chapter 8

Introduction to Topology

8.1 Why This Matters to Economists

This chapter and the associated quiz 5 on open and closed sets is intended for MRes students and MSc students who wish to take EC487 (Advanced Microeconomics) which is compulsory for MSc Econometrics and Mathematical Economics (EME) students and available to MSc Economics students with permission of the instructor. If you struggle unsuccessfully with the quiz and chapter you are likely to find EC487 very difficult. If you are not an MRes or MSc EME student and have no interest in EC487 do not attempt quiz 5 and read this chapter unless you are very confident that you have a good grasp of the rest of EC400 maths and enjoy the challenge.

This quiz covers the beginnings of real analysis in $\mathcal{R}^n$ and starts the move to topology. This is relatively abstract material. If you have done a course in real analysis this may be useful revision of the basics and might help to build your intuition. The concepts here are required for rigorous proofs of some important results in microeconomics.

The mathematics in this chapter matters to economists for several reasons. There is a big result, that a function that is continuous on a closed and bounded subset of $\mathcal{R}^n$ has a maximum and a minimum. This is important because maximization and minimization crop up very frequently in economics. You will meet the mathematics discussed in this chapter very early in a mathematically rigorous microeconomics course in the form of the continuity assumptions for consumer preferences and its implication, the existence of a continuous utility functions that captures the preferences. Continuity is also central to theorems on the existence of general competitive equilibrium.

Sections 1-4 are very useful. The rest of the chapter is harder going and less widely used in economics. Section 9 is the background you need to understand the material on continuity in consumer theory.

8.2 Vector Length and Open Balls

Remember from the chapter 2 that in $\mathcal{R}^n$ we have a concept of the length or norm of a vector

$$\|\mathbf{x}\| = \left(\sum_{i=1}^{n} x_i^2\right)^{\frac{1}{2}}.$$

In one dimensional space $n = 1$, and $\|x\| = |x|$, the absolute value of x. In two and three dimensional space Pythagoras' Theorem implies that this is indeed the length of the vector. The norm has the property that

$$\begin{aligned} \|\mathbf{x}\| &\geq 0 \text{ for all } \mathbf{x} && (8.1) \\ \|\mathbf{x}\| &= 0 \text{ if and only if } \mathbf{x} = 0. && (8.2) \end{aligned}$$

The distance between two vectors $\mathbf{x}$ and $\mathbf{x}_0$ in $\mathcal{R}^n$ is defined as

$$\|\mathbf{x} - \mathbf{x}_0\| = \left(\sum_{i=1}^{n} (x_i - x_{0i})^2\right)^{\frac{1}{2}}.$$

In one dimensional space this reduces to the absolute value $|x - x_0|$ which is the distance between the numbers x and x_0. In two and three dimensional space Pythagoras' Theorem implies that $\|\mathbf{x} - \mathbf{x}_0\|$ is the distance between $\mathbf{x}$ and $\mathbf{x}_0$. I therefore refer to $\|\mathbf{x} - \mathbf{x}_0\|$ as distance, even when $n > 3$. Properties 8.1 and 8.2 imply that

$$\begin{aligned} \|\mathbf{x} - \mathbf{x}_0\| &\geq 0 \text{ for all } \mathbf{x} \\ \|\mathbf{x} - \mathbf{x}_0\| &= 0 \text{ if and only if } \mathbf{x} = \mathbf{x}_0. \end{aligned}$$

Now think about the set of points that are closer to $\mathbf{x}_0$ than some strictly positive number δ. In one dimensional space $\mathbf{x}_0$ is a number and this is the open interval $(x_0 - \delta, x_0 + \delta)$. In two dimensional space it is a circle centred on $\mathbf{x}_0$ with radius δ. In three dimensional space this is a sphere or ball. In fact we use the term ball regardless of the dimension of the space, using the following definition.

Definition 48 *If δ is a strictly positive number and $\mathbf{x}_0$ is an element of $\mathcal{R}^n$ the **open ball** $B(\mathbf{x}_0, \delta)$ centred on $\mathbf{x}_0$ with radius δ is the set of points that are closer than δ to $\mathbf{x}_0$, that is*

$$B(\mathbf{x}_0, \delta) = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n, \|\mathbf{x} - \mathbf{x}_0\| < \delta\}.$$

Note that the inequality in this definition $<$ is strict. I have used the word "open" many times before when writing about "open intervals" and I will explain shortly what the word "open" means in general. In order to do this I will introduce some more terms that mathematicians have borrowed from elsewhere.

8.3 Boundaries

Imagine yourself standing on the equator. Take a tiny step to the north; you are in the northern hemisphere. Take a tiny step to the south; you are in the southern hemisphere. The equator is the boundary of the two hemispheres. Mathematicians have borrowed the term boundary from geographers and use it much the same sense, applying it to the n dimensional space of real number $\mathcal{R}^n$.

Definition 49 *A point* $\mathbf{x}_0$ *in* $\mathcal{R}^n$ *is an element of the* ***boundary*** *of a subset* S *of* $\mathcal{R}^n$ *if for any number* $\delta > 0$ *the open ball* $B(\mathbf{x}_0, \delta)$ *contains both points that are elements of* S *and points that are in the complement* S^C *of* S*.*

You may recall that the definition of complement:

Definition 50 *The complement* S^C *of a subset* S *of* $\mathcal{R}^n$ *is the set of elements of* $\mathcal{R}^n$ *that are not in* S*, that is*

$$S^C = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n,\, \mathbf{x} \notin S\}.$$

The fact that the radius δ of the open ball $B(\mathbf{x}_0, \boldsymbol{\delta})$ can be a very small positive number captures the idea that a step to the north, however tiny, moves you from the equator to the northern hemisphere. The definitions of the boundary and the complement of a set immediately imply a result that is useful later.

Proposition 51 *The boundary of a set* S *and the boundary of its complement* S^C *are the same.*

The idea of a boundary may become clearer with an example.

Example 52 *The boundary of the set*

$$S = \left\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2,\, x_1 > x_2\right\}$$

shown in figure 8.1 is the straight line $x_1 = x_2$*.*

This example is illustrated in Figure 8.1. Any open ball such as A centered on a point on the line $x_1 = x_2$ has points that are in S and points that are not in S, so lies in the boundary of S. However any open ball such as B centered on a point that lies in S does not contain any points that are not in S if its radius is small enough. Similarly an open ball such as C that is centered on a point that does not lie in S or on the line $x_1 = x_2$ does not contain any points that are in S if its radius is small enough. Thus points that are not on the line $x_1 = x_2$ are not on the boundary of S.

8.4 Open and Closed Sets and Boundaries

British suburban gardens usually have something physical marking the boundary, a fence, a hedge or a wall; you could describe them as closed. Suburban gardens in the USA often have nothing visible to mark the boundary; you could describe them as open. Mathematicians use the terms open and closed in a somewhat similar way.

- A subset S of $\mathcal{R}^n$ is **open** if **none** of the points in the boundary of S lie in S.

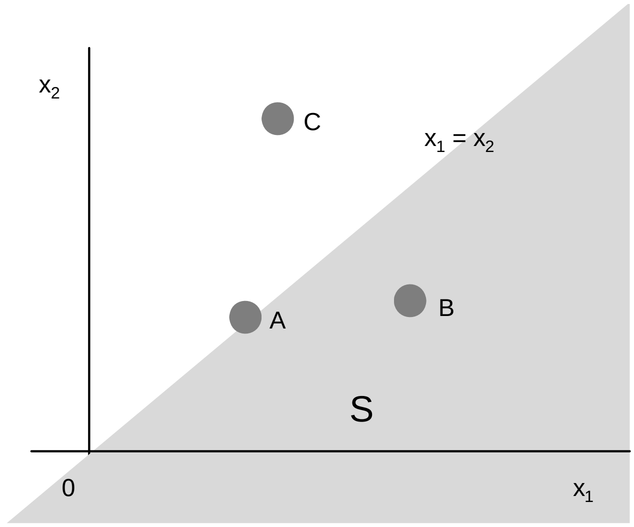

Figure 8.1: The set $\{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2,\, x_1 > x_2\}$

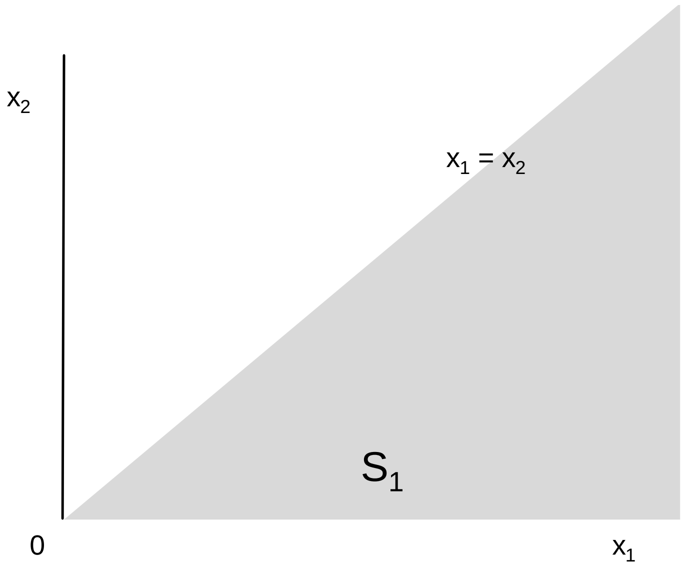

Figure 8.2: $S_1 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 < x_1\}$

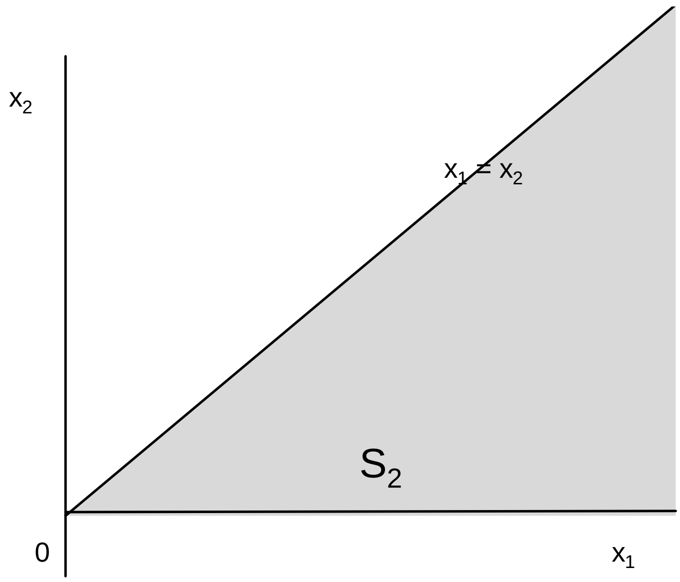

Figure 8.3: $S_2 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 \le x_2 \le x_1\}$

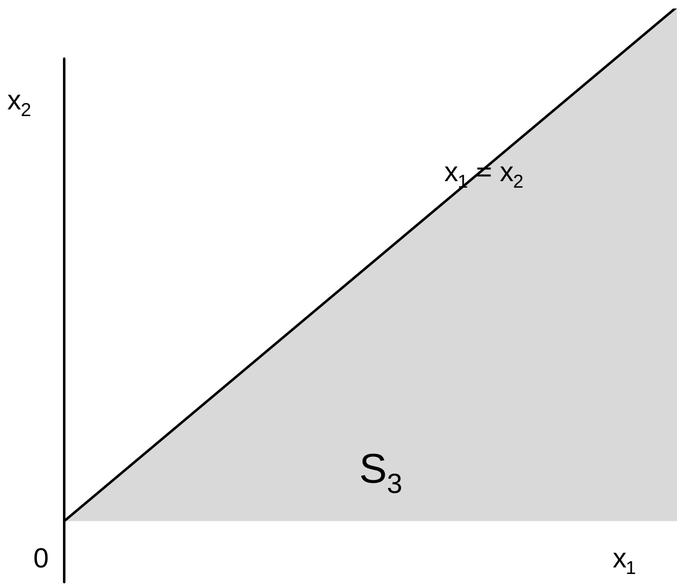

Figure 8.4: $S_3 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 \le x_1\}$

- A subset S of $\mathcal{R}^n$ is **closed** if **all** the points in the boundary of S lie in S.

For example the boundary of the set $S_1 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 < x_1\}$ in Figure 8.2 is the lines $x_2 = 0$ and $x_2 = x_1$ with $x_1 \geq 0$. These points are not in S_1 so S_1 is open. The boundary of the set $S_2 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 \leq x_2 \leq x_1\}$ in Figure 8.3 is the same as the boundary of S_1 but these points lie within S_2 so S_2 is closed. The lower boundary of the set $S_3 = \{(x_1, x_2) : (x_1, x_2) \in \mathcal{R}^2, 0 < x_2 \leq x_1\}$ in Figure 8.4 is not in S_3 but the upper boudary is in S_3. The set S_3 is neither open nor closed. You may notice that the set is closed if there all the inquality signs defining the sets are weak $\leq$ and open if all the inquality signs are strong $<$. This is general provided the functions used to define the set are continuous, but has not been proved at this stage.

8.5 Open Sets

8.5.1 Definition

The definition of an open set as a set for which none of the points in the boundary of S lie in S is good for an intuitive understanding of what an open set is. However this definition is not good for proving results about open sets, for which the "official" definition works better. This is:

Definition 53 *A subset S of $\mathcal{R}^n$ is **open** if for every element $\mathbf{x}$ of S there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S.*

The first result about open sets has to be that the "official" definition is equivalent to the definition in terms of the set boundary. This is:

Proposition 54 *A subset S of $\mathcal{R}^n$ is open if and only if it does not contain any points in its boundary.*

Proof. As S is open then for any element $\mathbf{x}$ of S there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S so does not contain any element of the complement S^C of S. Thus $\mathbf{x}$ cannot be a boundary point of S, so no points in S are in the boundary of S.

Now suppose that S does not contain any of its boundary points, but is not open. As S is not open there is an element $\mathbf{x}$ of S with the property that every open ball $B(\mathbf{x}, \delta)$ contains a point in S^C. As $\mathbf{x}$ is in S this implies that $\mathbf{x}$ is in the boundary of S, contradicting the assumption that S does not contain any of its boundary points. Thus if S does not contain any of its boundary points S must be open. ■

8.5.2 Open Intervals, Open Balls and Open Sets

I have used the term "open interval" frequently. I defined open intervals as sets of the form

$$\begin{aligned}
(a, b) &= \{x : x \in \mathcal{R}, a < x < b\} \\
(a, \infty) &= \{x : x \in \mathcal{R}, a < x\} \\
(-\infty, b) &= \{x : x \in \mathcal{R}, x < b\}.
\end{aligned}$$

The next proposition, which is proved in the appendix to this chapter establishes that this is a sensible form of words.

Proposition 55 *An open interval is an open subset of $\mathcal{R}$.*

The next bit of tidying up of the use of the word open is:

Proposition 56 *An open ball is an open set.*

Again this is proved in the appendix to this chapter.

8.5.3 Unions and Intersections of Open Sets

There are two important results here. The first is

Proposition 57 *The* ***union*** *of a* ***finite or infinite*** *number of open subsets of $\mathcal{R}^n$ is open.*

Proof. If $\mathbf{x}$ is an element of the union it is an element of one of the open sets S, so there is an open ball $B(\mathbf{x}, \delta)$ that is a subset of S, and thus a subset of the union of the sets. Thus the union is open. ■

The second result is:

Proposition 58 *The* ***intersection*** *of a* ***finite*** *number of open sets of $\mathcal{R}^n$ is open.*

Proof. Let the subsets be $S_1, S_2....S_n$, and let $\mathbf{x}$ be a point in their intersection. As S_i is open there is a number $\delta_i > 0$ such that the open ball $B(\mathbf{x}, \delta_i)$ is a subset of S_i, that is

$$B(\mathbf{x}, \delta_i) \subset S_i \tag{8.3}$$

Let

$$\delta = \min(\delta_1, \delta_2, ...\delta_n).$$

Thus $0 < \delta \leq \delta_i$ for all i so the open ball $B(\mathbf{x}, \delta)$ is a subset of the open ball $B(\mathbf{x}, \delta_i)$ for all i, that is $B(\mathbf{x}, \delta) \subset B(\mathbf{x}, \delta_i)$. Given relationship 8.3 this implies

$$B(\mathbf{x}, \delta) \subset S_i \text{ for } i = 1, 2...n$$

implying that $B(\mathbf{x}, \delta)$ is a subset of the intersection of $S_1, S_2....S_n$, so the intersection is itself open. ■

Note that the fact that this result applies to a **finite** number of open sets is important. For example that sets $B\left(\mathbf{0}, \frac{1}{n}\right)$ for $n = 1, 2, 3.....$ are open, but their intersection is $\{\mathbf{0}\}$ which is not an open set, because no open ball centred on $\mathbf{0}$ is a subset of $\{\mathbf{0}\}$.

8.6 Closed Sets

8.6.1 Formal Definition

I introduced closed sets as sets that include their boundaries. The "official" definition is

Definition 59 *A set is closed if its complement is open.*

The next proposition is that this definition is equivalent to the definition in terms of the set boundary.

Proposition 60 *A subset S of $\mathcal{R}^n$ is closed if and only if its boundary is a subset of S.*

Proof. Assume the set S is closed so its complement S^C is open. From Proposition 54 as S^C is open none of the points in its boundary lie in S^C, so all the points in the boundary of S^C lie in S. As the boundary of S and the boundary of its complement S^C are the same (Proposition 51) this implies that the boundary of the closed set S lies in S.

Conversely if the boundary of S lies in S there is no point in the boundary of S that lies in S^C. As the boundaries of S and S^C are the same this implies that no point in the boundary of S^C lies in S^C. From Proposition 54 this implies that S^C is an open set so S is a closed set. ■

8.6.2 Closed Sets and Infinite Sequences

You may have seen a different definition of a closed set in terms of limits of infinite sequences. The intuitive idea of the limit of an infinite sequence is that points in the infinite sequence get closer and closer to the limit as you move further and further along the sequence. The formal definition is:

Definition 61 *The infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ in $\mathcal{R}^n$ tends to a limit $\mathbf{x}_L$ in $\mathcal{R}^n$ if for any number $\varepsilon > 0$ there is an integer N such that $\|\mathbf{x}_i - \mathbf{x}_L\| < \varepsilon$, or equivalently $\mathbf{x}_i \in B(\mathbf{x}_L, \varepsilon)$, for all $i > N$.*

The result linking infinite sequences and closed sets is:

Proposition 62 *A subset S of $\mathcal{R}^n$ is closed if and only if, for any infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ whose points are all S that tends to a limit, that limit is in S.*

The proof of this proposition is in the appendix to this chapter. This result shows that I could have defined closed sets in $\mathcal{R}^n$ in terms of infinite sequences, as is sometimes done. However the definition of closed sets as sets whose complements are open is easier to work with.

8.7 Continuous functions

8.7.1 Continuous Functions of a Single Variable

Intuitively a function is continuous if $f(\mathbf{x})$ is very close to $f(\mathbf{x}_0)$ when $\mathbf{x}$ is very close to $\mathbf{x}_0$. Informally a function of a single variable is continuous if you can draw its graph without taking your pen off the paper. The function illustrated in Figure 8.5 is not continuous, the function in Figure 8.6 is continuous. The formal definition of continuity for a function of a single variable defined on an open[1] set is:

[1] At this point I work with a function defined on an open set S because then if $\mathbf{x}$ is in the set any open ball $B(\mathbf{x}, \delta)$ with a small enough δ lies entirely in the set. If the set is not open there are boundary points $\mathbf{x}$ in S for which $B(\mathbf{x}, \delta)$ is not a subset of S for any δ, and the function may not be defined at some points in the open ball.

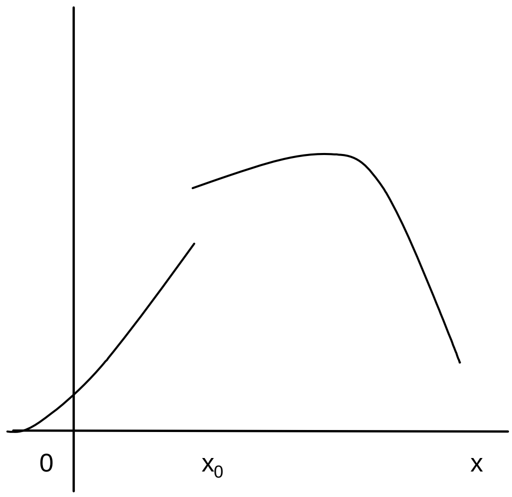

Figure 8.5: A function that is not continuous at x_0

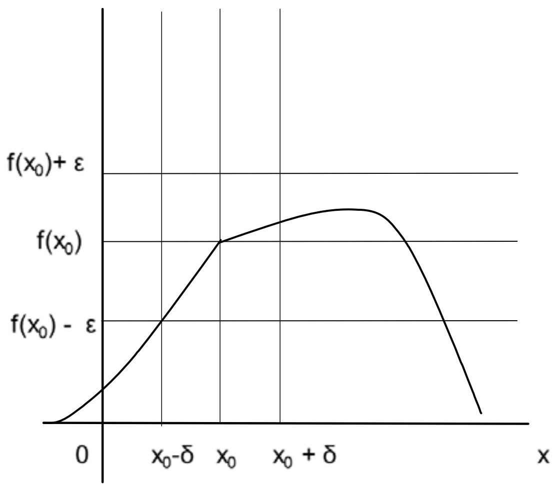

Figure 8.6: A function that is continous at x_0.

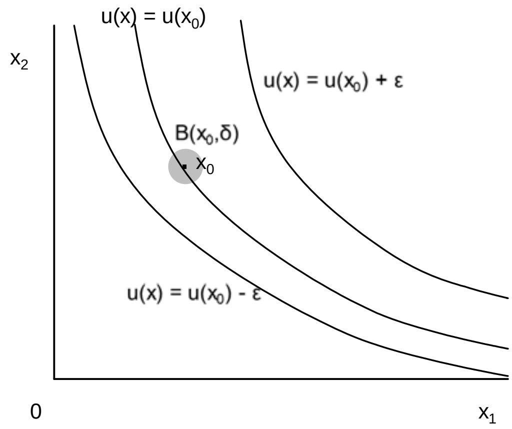

Figure 8.7: Indifference curves for a continuous utility function

Definition 63 *If S is an open subset of $\mathcal{R}$ the function $f : S \to \mathcal{R}$ is **continuous** if for any $x_0 \in \mathcal{S}$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.*

Figure 8.6 illustrates this formal definition; if x is within a distance δ of x_0 then $f(x)$ is within a distance ε of $f(x_0)$. Note that because the graph of the function in figure 8.6 has a kink at x_0 the function does not have a derivative. Continuity and differentiability are different concetps.

8.7.2 Continuous Functions of Several Variables

The notation for a function of n variables is $f(\mathbf{x})$ where $\mathbf{x}$ is an n vector, that is an element of $\mathcal{R}^n$. The intuitive idea of continuity for functions of several variables is the same as the intuition for functions of a single variable; if $\mathbf{x}$ is very close to $\mathbf{x}_0$ then $f(\mathbf{x})$ is very close to $f(\mathbf{x}_0)$. The formal definition of continuity for functions of several variables requires cutting and pasting the definition for functions of a single variable, replacing $\mathcal{R}$ by $\mathcal{R}^n$ where appropriate, $|x - x_0|$ by $\|\mathbf{x} - \mathbf{x}_0\|$ and $f(x) - f(x_0)$ by $f(\mathbf{x}) - f(\mathbf{x}_0)$.

Definition 64 (Continuity for a Function of Several Variables 1) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for any $\mathbf{x}_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\|\mathbf{x} - \mathbf{x}_0\| < \delta$, then $|f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon$.*

It is somewhat easier to visualize with an equivalent definition

Definition 65 (Continuity for a Function of Several Variables 2) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for any $\mathbf{x}_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\mathbf{x} \in B(\mathbf{x}_0, \delta)$ then $|f(\mathbf{x}_0) - f(\mathbf{x})| < \delta$*

To see that the two definitions are equivalent observe that the set of values of $\mathbf{x}$ for which $\|\mathbf{x} - \mathbf{x}_0\| < \delta$ is the open ball $B(\mathbf{x}_0, \delta)$.

Figure 8.7 illustrates this definition for one of economists' favorite functions and diagrams, a utility function and indifference curves. There is an open ball $B(\mathbf{x}_0, \delta)$ that lies in the set between the two indifference curves with $u(\mathbf{x}) = u(\mathbf{x}_0) - \varepsilon$ and $u(\mathbf{x}) = u(\mathbf{x}_0) - \varepsilon$. For all points in this open ball $|u(\mathbf{x}) - u(\mathbf{x}_0)| < \varepsilon$.

8.8 Closed Sets, Bounded Sets, Compact Sets and Continuous Functions

For economists one of the most important results on continuous functions is:

Theorem 66 *If S is a* ***closed and bounded*** *subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$ is* ***continuous*** *the function* ***has a maximum and a minimum*** *on S, that is there are elements $\mathbf{x}_{\min}$ and $\mathbf{x}_{\max}$ of S with the property that*

$$f(\mathbf{x}_{\min}) \leq f(\mathbf{x}) \leq f(\mathbf{x}_{\max})$$

for all $\mathbf{x}$ in S.

The theorem is important due to economists' obsession with maximization as it gives conditions under which you can be sure that the function has a maximum and minimum. The proof is somewhat complicated and I do not give it here.

This definition is no use without a definition of a bounded set:

Definition 67 *A subset S of $\mathcal{R}^n$ is* ***bounded*** *if there is an element y of $\mathcal{R}$ with the property that*

$$\|\mathbf{x}\| < y \text{ for all } \mathbf{x} \text{ in } S.$$

Another way of saying this is that the distance between $\mathbf{x}$ and $\mathbf{0}$ is less than y for all elements of S. If $n = 1$ this simply means that $-y < x < y$ for all x in S. If $n = 2$ this means that S lies inside a circle with radius y centered on the origin. More generally it means that all $\mathbf{x}$ in S lie in the open ball $B(\mathbf{0}, y)$.

It is important that S be both closed and bounded. For example the interval $(0, 1)$ is bounded but not closed, the function $f(x) = x^{-1}$ is continuous on $(0, 1)$ but has no maximum, as $f(x)$ grows without limit as x gets closer and closer to 0. The interval $[0, \infty)$ is closed but not bounded, and the continuous function $f(x) = x^2$ grows without limit as x tends to infinity.

If you have done some general topology you will have come across the term "compact". There is a general definition of a compact set and then a theorem (the Heine-Borel Theorem) which states that the sets in $\mathcal{R}^n$ that are compact are the sets that are closed and bounded, so when working in $\mathcal{R}^n$ you should interpret compact as meaning closed and bounded. If you have never done any general topology ignore this comment.

8.9 Continuity for Consumer Theory

8.9.1 The Definition

Continuity is important at one point in consumer theory, showing that the continuity assumption on preferences implies the existence of a continuous utility function that represents the preferences. In order to understand this result you need to know that continuity can be defined in the following way.

Definition 68 (Continuity for a Function of Several Variables 3) *If S is a subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is* ***continuous*** *if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

For this to make sense this definition has to be equivalent to Definitions 64 and 65 of continuous functions. The rest of the chapter demonstrates that this is indeed so. This is very much an exercise in pure mathematics with a number of real analysis type arguments with ε and δ. I suggest skipping this section if you have never seen any real analysis.

8.9.2 Level Sets, Upper Contour Sets, Lower Contour Sets

It is useful to have some terminology.

Definition 69 *If S is a subset of $\mathcal{R}$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) = y\}$$

of elements of S for which $f(\mathbf{x}) = y$ is called a ***level set****.*

This may be a new piece of vocabulary, but economists are very familiar with level sets; we call the level set of a utility function an indifference curve, and the level set of a production function an isoquant.

Definition 70 *If S is a subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$$

of elements of S for which $f(\mathbf{x})$ is greater than y is called an ***upper contour set****.*

Definition 71 *If S is a subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$$

of elements of S for which $f(\mathbf{x})$ is less than y is called a ***lower contour set****.*

Figure 8.8 illustrates these sets for a utility function.

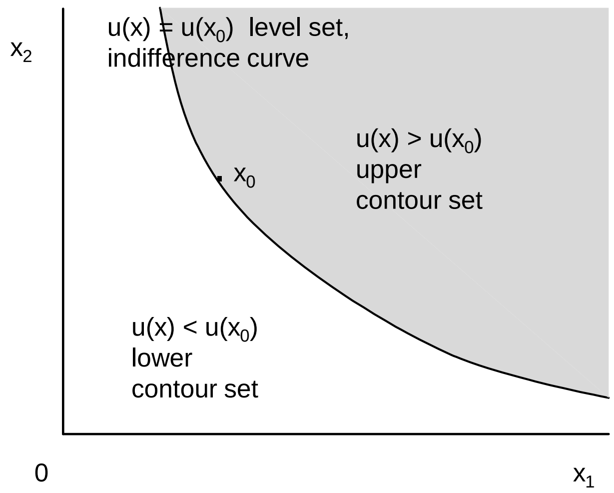

Figure 8.8: Level set, upper contour set and lower contour set for a utility function

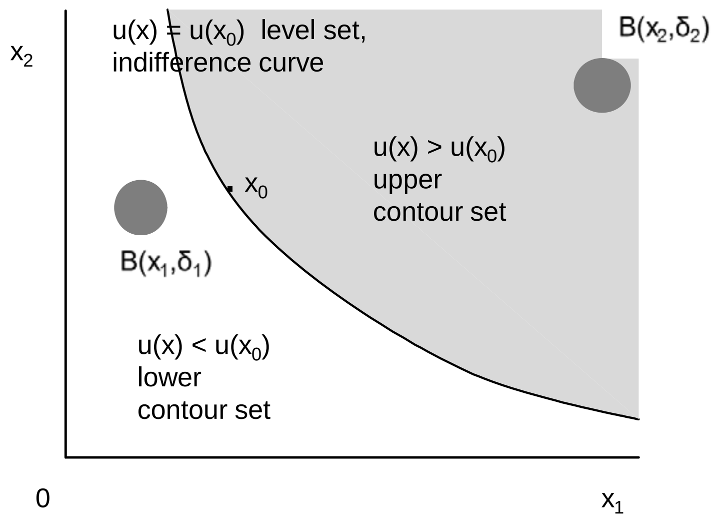

Figure 8.9: Open upper and lower contour sets

8.9.3 Open Set and Closed Set Definition of Continuity

The argument that Definition 68 of continuity is equivalent to Definitions 64 and 65 starts from the following proposition.

Proposition 72 *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is continuous if and only if for all elements y of $\mathcal{R}$ the upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$ and the lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$ are both open.*

Figure 8.9 illustrates the conditions of this thereon for a utility function. As the lower contour set is open, for any point $\mathbf{x}_1$ in the lower contour set, there is an open ball $B(\mathbf{x}_1, \delta_1)$ that is a subset of the lower contour set. As the upper contour set is open, for any point $\mathbf{x}_2$ in the upper contour set, there is an open ball $B(\mathbf{x}_2, \delta_2)$ that is a subset of the upper contour set.

The proof of Proposition 72 is suggested by Figures 8.8 and 8.9.

Proof. The function $f : S \to \mathcal{R}$ is continuous if for any $x_0 \in S$ and any $\varepsilon > 0$ there is a number $\delta > 0$ with the property that if $\mathbf{x} \in B(\mathbf{x}_0, \delta)$ then $|f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon$. In particular if $f(\mathbf{x}_1) < y$ so $y - f(\mathbf{x}_1) > 0$ there is an open ball $B(\mathbf{x}_1, \delta_1)$ on which $|f(\mathbf{x}) - f(\mathbf{x}_1)| < y - f(\mathbf{x}_1)$. As $f(\mathbf{x}) - f(\mathbf{x}_1) \leq |f(\mathbf{x}) - f(\mathbf{x}_1)|$ which implies that $f(\mathbf{x}) - f(\mathbf{x}_1) < y - f(\mathbf{x}_1)$ so $f(\mathbf{x}) < y$. Thus the lower contour set is open. Similarly if $f(\mathbf{x}_2) > y$ so $f(\mathbf{x}_2) - y > 0$ there is an open ball $B(\mathbf{x}_2, \delta_2)$ on which $|f(\mathbf{x}_2) - f(\mathbf{x})| < f(\mathbf{x}_2) - y$. As $f(\mathbf{x}_2) - f(\mathbf{x}) \leq |f(\mathbf{x}_2) - f(\mathbf{x})|$ this implies that $f(\mathbf{x}_2) - f(\mathbf{x}) < f(\mathbf{x}_2) - y$ so $y < f(\mathbf{x})$. Thus the upper contour set is open. Hence the upper and lower contour sets are open if the function is continuous.

To prove the converse result note that if the upper contour set

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}_0) - \varepsilon < f(\mathbf{x})\}$$

and the lower contour set

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < f(\mathbf{x}_0) + \varepsilon\}$$

are open then, from Proposition 58, their intersection is also open. This intersection is the set on which $f(\mathbf{x}_0) - \varepsilon < f(\mathbf{x}) < f(\mathbf{x}_0) + \varepsilon$, or equivalently the set on which $-\varepsilon < f(\mathbf{x}) - f(\mathbf{x}_0) < \varepsilon$, that is the set $\{\mathbf{x} : \mathbf{x} \in S,\ |f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon\}$. The point $\mathbf{x}_0$ is an element of this set, hence as the set is open there is an open ball $B(\mathbf{x}_0, \delta)$ with the property that all elements of the open ball lie in the set $\{\mathbf{x} : \mathbf{x} \in S,\ |f(\mathbf{x}) - f(\mathbf{x}_0)| < \varepsilon\}$. This is what is needed to establish continuity. ■

This proposition makes possible yet another equivalent definition of continuity.

Definition 73 (Continuity for a Function of Several Variables 4) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is* ***continuous*** *if for all elements y of $\mathcal{R}$ the upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$ and the lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$ are both open.*

Now suppose that $f(\mathbf{x}) : S \to \mathcal{R}$ is continuous and consider the complement in S of the open upper contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) > y\}$, that is the set

$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$. This is by definition a closed set[2] . Similarly the complement in S of the open lower contour set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) < y\}$, that is the closed set $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$. Given Definition 73 I can give yet another definition of continuity.

Definition 74 (Continuity for a Function of Several Variables 5) *If S is an open subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

I have been assuming throughout this argument that S is an open set. With a little more mathematical sophistication[3] I can drop the requirement that S is an open set. This is useful thing to do, because for may purposes it is necessary to think of functions defined on $\mathcal{R}^{n+}$ which is defined as the set of vectors with $x_i \geq 0$ for $i = 1, 2...n$. The set $\mathcal{R}^{n+}$ is closed.

Definition 75 (Continuity for a Function of Several Variables) *If S is a subset of $\mathcal{R}^n$ the function $f : S \to \mathcal{R}$ is **continuous** if for all elements y of $\mathcal{R}$ the sets $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \leq y\}$ and $\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) \geq y\}$ are both closed.*

This is the definition used in consumer theory.

8.10 Appendix: Proofs

8.10.1 Proof That An Open Interval is an 0pen Subset of $\mathcal{R}$

If $a < b$, $a < x < b$ and $\varepsilon = \min(x - a, b - x)$ then as $a < x < b$

$$\varepsilon > 0. \tag{8.4}$$

As $\varepsilon = \min(x - a, b - x) \leq x - a$ and $\varepsilon > 0$

$$a \leq x - \varepsilon < x - \frac{1}{2}\varepsilon < x$$

so

$$a < x - \frac{1}{2}\varepsilon < x < b. \tag{8.5}$$

As $\varepsilon = \min(x - a, b - x) \leq b - x$

$$b \geq x + \varepsilon > x + \frac{1}{2}\varepsilon > x > a$$

[2] I am being somewhat sloppy with definitions of open and closed sets here. If you have not noticed my sloppiness ignore this footnote. The difficulty is that I am assuming that S is open and f is continuous, in which case the set $\{\mathbf{x} : \mathbf{x} \in S, f(\mathbf{x}) \leq y\}$ that I am calling the complement of the upper contour set in S is not a closed subset of $\mathcal{R}^n$ unless $S = \mathcal{R}^n$. I can get away with this by defining an open set in S as the intersection of S and a set that is open in $\mathcal{R}^n$, and a closed set in S as the complement in S of an open set in S. If S is open the open sets in S are open in $\mathcal{R}^n$, but the closed sets in S are not closed in $\mathcal{R}^n$. If S is closed some of the open sets in S are not open in $\mathcal{R}^n$, but the closed sets in S are closed in $\mathcal{R}^n$. If you know about general topology you will recognize that either way I am setting up a topology on S, and know that this makes mathematical sense.

[3] In order to do this I have to define an open set in S as the intersection of an open set in $\mathcal{R}^n$, and then define a closed subset of S as the complement in S of an open subset in S. See footnote 2.

so

$$a < x < x + \frac{1}{2}\varepsilon. \tag{8.6}$$

Taken together inequalities 8.4, 8.5 and 8.6 imply that

$$a < x - \frac{1}{2}\varepsilon < x < x + \frac{1}{2}\varepsilon < b. \tag{8.7}$$

The set of points satisfying $x - \frac{1}{2}\varepsilon < x < x + \frac{1}{2}\varepsilon$ is the open ball $B\left(x, \frac{\varepsilon}{2}\right)$. Inequality 8.7 then implies that $B\left(x, \frac{\varepsilon}{2}\right)$ is a subset of the intervals (a, b) (a, ∞) and $(-\infty, b)$. Thus these intervals are open.

8.10.2 Proof That An Open Ball is an Open Subset of $\mathcal{R}^n$

The open ball $B(\mathbf{x}_0, \delta)$ is the set of points satisfying $\|\mathbf{x} - \mathbf{x}_0\| < \delta$. Note that if $\mathbf{x}$ is an element of $B(\mathbf{x}_0, \delta)$ then $\delta - \|\mathbf{x} - \mathbf{x}_0\| > 0$. Let $\mathbf{z}$ be an element of the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$. Then

$$\|\mathbf{z} - \mathbf{x}\| < \delta - \|\mathbf{x} - \mathbf{x}_0\|.$$

From the triangle inequality (proved in the previous chapter on vectors)

$$\|\mathbf{z} - \mathbf{x}_0\| = \|\mathbf{z} - \mathbf{x} + \mathbf{x} - \mathbf{x}_0\| \leq \|\mathbf{z} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\|.$$

The last two inequalities imply that

$$\|\mathbf{z} - \mathbf{x}_0\| \leq \|\mathbf{z} - \mathbf{x}\| + \|\mathbf{x} - \mathbf{x}_0\| < \delta - \|\mathbf{x} - \mathbf{x}_0\| + \|\mathbf{x} - \mathbf{x}_0\| = \delta.$$

Thus if $\mathbf{z}$ is an element of the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$ then $\|\mathbf{z} - \mathbf{x}_0\| < \delta$, so $\mathbf{z}$ is an element of the open ball $B(\mathbf{x}_0, \delta)$. Thus the open ball $B(\mathbf{x}, \delta - \|\mathbf{x} - \mathbf{x}_0\|)$ is a subset of $B(\mathbf{x}_0, \delta)$, so the open ball $B(\mathbf{x}_0, \delta)$ is an open set.

8.10.3 Infinite Sequences and Closed Sets

The result here is

Proposition 76 *A set S is closed if and only if for any infinite sequence $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ whose points are all S that tends to a limit, that limit is in S.*

Proof. Suppose that S is closed and that $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ is an infinite sequence of points in S that tend to a limit in S^C. As S is closed S^C is open so there is an open ball $B(\mathbf{x}_L, \varepsilon)$ that is a subset of S^C. As $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ tends to $\mathbf{x}_L$ the point $\mathbf{x}_i$ is an element of $B(\mathbf{x}_L, \varepsilon)$ and thus of S^C for all sufficiently large i. But by assumption all points in the sequence lie in S so cannot be in S^C. This contradiction implies that the limit of infinite sequence of points a closed set S must lie in S.

Now suppose that the limit of any infinite sequence of points in S lies in S, but S is not closed. As S is not closed, S^C is not open. This implies that there is a point $\mathbf{x}$ in S^C for which every open ball $B(\mathbf{x}, \varepsilon)$ contains a point that is not in S^C and therefore is in S, regardless of how small ε is. Thus there is an infinite sequence of points $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3.......\}$ that are not in S^C for which $\mathbf{x}_i \in B\left(\mathbf{x}, \frac{1}{i}\right)$. This infinite sequence tends to $\mathbf{x}$. As none of the points in the infinite sequence are in S^C they are all in S. However the limit is in S^C. This contradicts the supposition on limits of infinite sequences and S not being closed that I started with. Thus S is closed. ■